SOFTWARE REQUIREMENTS SPECIFICATION

for

Secure and Reliable Residential Elevator Monitoring and Control System

SREMS | Version 1.1 - Final Submission

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1 Introduction

This Software Requirements Specification (SRS) defines the expected behavior of the Secure and Reliable Residential Elevator Monitoring and Control System (SREMS) in a 30-floor residential building with one basement parking level. It is written to guide implementation and testing and the requirements emphasize two product qualities: security and reliability.

1.1 Purpose

The purpose of this document is to establish a shared, testable description of what SREMS must provide to passengers, building owners, security managers, and technicians. It covers passenger service, elevator monitoring, floor-level authorization, security-event handling, emergency response, fault isolation, maintenance support, auditing, and analytics. Basic controller behavior (button handling, car movement) is assumed and covered only minimally.

Security means that protected destinations and control actions require appropriate permission; attempts and consequential actions are traceable; and malformed external messages cannot silently change the system state. Reliability means that requests and state transitions are handled predictably, unavailable cars are excluded from normal assignment, critical events are recorded, and communication failures do not cause the system to act on stale information.

1.2 Scope

The building contains 30 above-ground floors (1-30) and B1, the basement parking level. It has four residential elevators (E1-E4) and one larger freight/service elevator (E5) that residents may also use to carry freight, goods, or pets. E1-E4 serve B1; E5 does not serve B1 under the current project scope.

SREMS shall provide:

- Fleet monitoring for current floor, direction, operational state, door state, request status, fault status, maintenance status, and emergency status when the elevator interface supplies the relevant data.
- Hall-call handling and direction-aware assignment of an eligible elevator.
- Destination selection with simulated card identity and configurable floor-level access control. Hall calls and travel to B1 do not require a card.
- A shared operational-panel concept for passenger controls and authorized technician actions, plus a role-restricted monitoring dashboard for building owners and security managers.
- Emergency and fault handling, including occupant-facing status, security notification, incident tracking, and explicit restrictions on normal dispatch.
- Maintenance isolation, diagnostics, maintenance history, security audit records, tamper-evidence verification, and service-performance analytics.
- Exported measurements for evaluating dispatch and reliability behavior, including optional repeatable benchmark runs (Section 3.3.4).
- Credential lifecycle management, including lost-card, cloned-card, and visitor-pass handling.
- Medical emergency handling and detection of call abuse (empty calls, repeated calls, and repeated emergency triggers).
- Freight/service runs on E5 for authorized building-service staff and technicians, and resident use of E5 for freight, goods, and pets when no run is active.

SREMS is software that receives elevator events and issues commands through a defined elevator interface (Section 3.3), so it can be driven by elevator equipment or by test stubs. It is not a certified physical elevator controller, and the project shall not claim that it guarantees real-world elevator safety. Physical sensors, door mechanisms, card readers, emergency phones, and public-address hardware are represented by elevator-interface events or software interfaces unless separately provided. The ESRA elevator simulator is an optional benchmarking tool (Section 3.3.4); no other requirement depends on it.

1.3 Definitions, Acronyms, and Abbreviations

Table 1: Definitions, acronyms and abbreviations

Term	Definition
Admin	A user holding the Security manager or Building owner role. Admins have the same authority in SREMS; the two roles differ only in dashboard focus (security events versus building operations).
B1	Basement parking level served by residential elevators E1-E4.

Term	Definition
Blocking fault	A reported fault after which SREMS keeps the car out of ordinary assignment until the recovery checks pass (MNT-4.1, MNT-4.6).
Building-emergency policy	The default rule set in EMR-3.6, applied to all cars in a fire or a power failure.
Building-service staff	Staff, such as cleaners or delivery handlers, whose credential allows them to start E5 service runs.
Car call	A destination request made from inside an elevator car.
Common floor	Floors, including Floor 1 and B1 for this specification, that every resident credential may select and travel to (SEC-2.3).
Credential	A card identifier linked to a user identity and a permission set. A credential is ACTIVE, SUSPENDED, REVOKED, or EXPIRED (SEC-2.16). A lost card is first SUSPENDED, then reinstated or REVOKED (SEC-2.17).
Elevator interface	The single logical boundary through which SREMS receives equipment events and issues commands (Section 3.3). Elevator equipment or test stubs sit behind it.
Elevator travel time	Elapsed time between a recorded departure from one floor and arrival at the requested destination, where the elevator interface provides enough timestamps.
Eligible elevator	A car permitted to receive a new ordinary assignment: operational, not under maintenance, not in an emergency/blocking-fault state, not OCCUPIED by a service run (FR-1.13), and has reported a valid state within the last 5 seconds (configurable).
Emergency state	The car state EMERGENCY/UNAVAILABLE, in which a car receives no new ordinary assignments (EMR-3.2). It is a software state and makes no physical safety claim.
ESRA	Elevator Simulation, Research & Analysis, an open-source simulator described by Tartan and Çiflikli (2024), used only for optional benchmarking.
Fail closed	On a failure, deny the restricted action instead of allowing it (MNT-4.11).
Hall call	A request made at a floor. An ordinary hall call asks for a car travelling up or down and is served by E1 to E4 only. A freight call asks for E5, is for freight, goods, or pets, and may be made by residents and staff. A hall call does not need to be authorized.
Incident	A tracked fault, emergency, or security event that requires review or action.
Maintenance mode	A car state set by a Technician or Admin, in which the car receives no new ordinary assignments until it is returned to service (MNT-4.3, MNT-4.6).
NO_BOARDING	A flag on a completed hall request when no boarding evidence follows the arrival of the car (FR-1.21).
Normal test conditions	A test run in which all five cars report state, the data store is healthy, and no more than 100 requests are active (CAP-1).
Passenger	Any person using an elevator, with or without a credential.
RBAC	Role-Based Access Control, which grants permissions according to the assigned role.
Recovery checks	The four checks listed in MNT-4.6, which must pass before a car returns to service.
Rescue car	An eligible car sent by an Admin to the floor of a car in an emergency so that its occupants can be reached, or an ill person can be picked up, and held there for an Admin-set wait time (EMR-3.12, EMR-3.16).

Term	Definition
Resident	A passenger whose credential is linked to an apartment floor.
Responder floor	The floor where a car in a medical emergency stops and opens its doors. The default is Floor 1 (EMR-3.14).
Security event	An authentication, authorization, command-validation, emergency, or audit event relevant to protecting the system.
Sensor	Equipment outside SREMS that reports a physical value through the elevator interface (HW-1).
SREMS	The elevator management software system specified in this SRS, providing passenger service, elevator monitoring, access control, security and emergency handling, fault and maintenance management, auditing, and analytics through a defined elevator interface.
Stale state	A displayed/held elevator state that cannot be confirmed as current because fresh elevator-interface data has not been received.
Tamper-evident log	An audit log whose integrity can be checked to detect unauthorized modification after records are created.
Test stub	A software stand-in for elevator equipment that supplies events or accepts commands through the elevator interface. Stub input is always labelled as such (SWI-2).
Traffic profile	A named set of request rates, peak or off-peak, used to test the wait-time targets (PERF-8, OR-10).
Wait time	Elapsed time from an accepted hall call to the assigned elevator arriving at the requesting floor.

1.4 References

The principal sources are listed in Section 5. The requirements use the general SRS structure of ISO/IEC/IEEE 29148:2018 and cite the ESRA paper only for the optional benchmark.

1.5 Overview

Section 2 describes the system context, functions, stakeholders, constraints, assumptions, and dependencies. Section 3 defines the external interfaces, functional requirements, logical elevator-interface contract, performance requirements, design constraints, and quality attributes. Section 4 identifies decisions that remain open and specifies how the team will validate the requirements during implementation. Section 5 lists references.

2 Overall Description

2.1 Product Perspective

SREMS is a software layer between its users and the elevator equipment. It manages requests, access policies, monitoring, security events, maintenance state, and records. Elevator cars, door units, load

sensors, and building alarms are outside SREMS and communicate with it only through the elevator interface (Section 3.3), so core behaviors can be tested without physical hardware.

This is a logical view, not a mandate to use a particular programming language, database, framework, or deployment topology. Internal module boundaries are design choices, but SREMS shall keep equipment-specific integration inside the elevator interface so that core behaviors can be tested independently.

2.2 Product Functions

- Passenger service: receive hall calls, select a suitable eligible elevator, display approach/status, accept authorized car calls, and record request outcomes.
- Fleet monitoring: show elevator position, direction, door state, availability, active requests, faults, maintenance, and emergency status when those data are available.
- Access control: simulate card credentials, associate identities with roles and floor permissions, reject unauthorized destination commands, and audit security-sensitive actions.
- Emergency management: receive emergency triggers, apply the defined project policy, support emergency overrides including rescue cars, notify security personnel and affected occupants through available software interfaces, and maintain an incident record.
- Reliability management: isolate cars under maintenance/blocking fault, detect stale state and lost elevator-interface communication, retain request/event records, and reconcile state after reconnecting.
- Maintenance: show current state and recent calls, allow authorized maintenance isolation and return-to-service operations, and record maintenance actions.
- Analytics: calculate wait time, travel time where measurable, utilization, fault/downtime, and security-event metrics, verify audit-log integrity, and optionally run repeatable benchmark profiles.
- Accessibility: present status through readable text and non-visual announcements for floor, direction, arrival, and emergency information.
- Freight/service operation: run E5 service trips for authorized building-service staff and technicians without stopping for other calls. E5 is available to residents at all other times, for freight, goods, and pets.
- Credential management: issue, replace, suspend, and revoke credentials, issue visitor passes, and detect lost, duplicated, and unregistered cards.
- Abuse detection: merge duplicate calls, flag empty calls, and alert on repeated emergency triggers.
- Performance targets: average wait-time targets under named peak and off-peak traffic profiles.

Table 2 lists the use cases shown in Figure 1. Behavior that is included in or extends a use case (verifying user authority, incident response, notifying security and recording an incident) is specified in UC-08, UC-10, and UC-11.

Table 2: Use cases

ID	Use case	Expected outcome
UC-01	Call an elevator	The passenger's hall call is submitted without any authentication check (FR-1.3, SEC-2.4). An ordinary call is served by E1 to E4 only, and a freight call by E5 only (FR-1.13). Repeated empty calls from one floor may raise a security alert (UC-16). The call is recorded and assigned to the best eligible elevator according to the configured direction-aware dispatch rule.
UC-02	Select destination	A passenger selects a destination inside the elevator. Where the destination requires authorization, the passenger is authenticated and the destination

ID	Use case	Expected outcome
		command is accepted only if the passenger has the required permission. B1 is treated as the defined no-card exception.
UC-03	Handle denied access	The command is rejected, the passenger receives feedback, and the event is recorded; repeated failures may create an alert.
UC-04	Trigger an emergency	A passenger or staff triggers an emergency, help, or medical event, and no card or permission is needed. A stuck event reported by the equipment has the same effect as a confirmed help request. For a medical or stuck-car event, the affected car enters the emergency state and is excluded from ordinary dispatch; for a help request, only the alert and the incident apply until the request is confirmed (EMR-3.1). Security is notified immediately, and an incident is recorded (see Incident response, UC-10). Repeated triggers on one car may raise a security alert (UC-16).
UC-05	Isolate/restore a faulted car	An elevator affected by a blocking fault or maintenance condition is removed from the ordinary dispatch pool. The elevator may be returned to normal service only by an authorized technician or Admin after the recovery checks in MNT-4.6 have passed.
UC-06	Review fleet/security	Authorized staff can inspect current elevator states, active requests, faults, incidents, maintenance information, security events, audit records, downtime, and available metrics. Admins can also run an audit-integrity check.
UC-07	Emergency override	An authenticated Admin issues an override permitted by the emergency policy, including sending a rescue car and deciding the waiting time. The command, user, time, and outcome are recorded in the audit log.
UC-08	Verify user authority	Before any restricted action, the system checks the user's card identity and role. An unauthorized attempt is rejected, which leads to Handle denied access (UC-03). Included by UC-05, UC-06, UC-07, UC-12 and UC-13; extends UC-02.
UC-09	Recover from communication loss	When communication with the elevator equipment is lost, the affected state is marked stale, no new control commands are sent, and staff are notified. After reconnection, a fresh snapshot is obtained and requests are reconciled before control resumes (MNT-4.8, MNT-4.9). Includes Incident response (UC-10).
UC-10	Incident response	After an emergency, a communication loss, a building-wide emergency, or an escalated sensor event, an incident is created and security is notified (UC-11). An Admin or Technician acknowledges and resolves it (EMR-3.9). If it is unresolved after the configured threshold, security is notified again and the escalation is recorded. Included by UC-04, UC-09 and UC-14; extends UC-15.
UC-11	Notify security and record incident	The dashboard shows the alert within 1 second (PERF-3), the occupant message is shown when a car is involved, and the incident record with its context is stored (EMR-3.4, EMR-3.7). Included by UC-10 and UC-16.
UC-12	Run a service trip (E5)	Authorized building-service staff or a Technician authenticate (UC-08) and start an E5 service run from the service control. A service run is not a freight call. The run goes directly to its destination without stopping for other calls, and E5 shows OCCUPIED until the run ends (FR-1.13). Resident credentials cannot start a run.
UC-13	Manage credentials	An Admin issues, replaces, suspends, reinstates, or revokes a credential, or issues a visitor pass, after authentication (UC-08). A lost card is suspended first and revoked when the review window ends. Each action is audited (SEC-2.16, SEC-2.17).

ID	Use case	Expected outcome
UC-14	Handle building-wide emergency	A fire alarm or power failure event arrives. The default building-emergency policy (EMR-3.6) is applied to all cars and an incident is created (UC-10).
UC-15	Handle sensor event or fault	A sensor event, such as a door obstruction or an overload, or a sensor fault is received. An obstruction beyond the threshold escalates to security (FR-1.15). A faulty sensor is marked FAULTY and staff are notified (HW-2). Escalation creates an incident (UC-10).
UC-16	Raise security alert	A repeated pattern is detected: three failed attempts on one credential (SEC-2.7), five empty calls from one floor (FR-1.22), more than three emergency triggers on one car (EMR-3.15), use of a suspended (lost), revoked, or cloned credential (SEC-2.17, SEC-2.18), or suspected card scanning (SEC-2.19). An alert is created and security is notified (UC-11). Extends UC-01, UC-03 and UC-04.

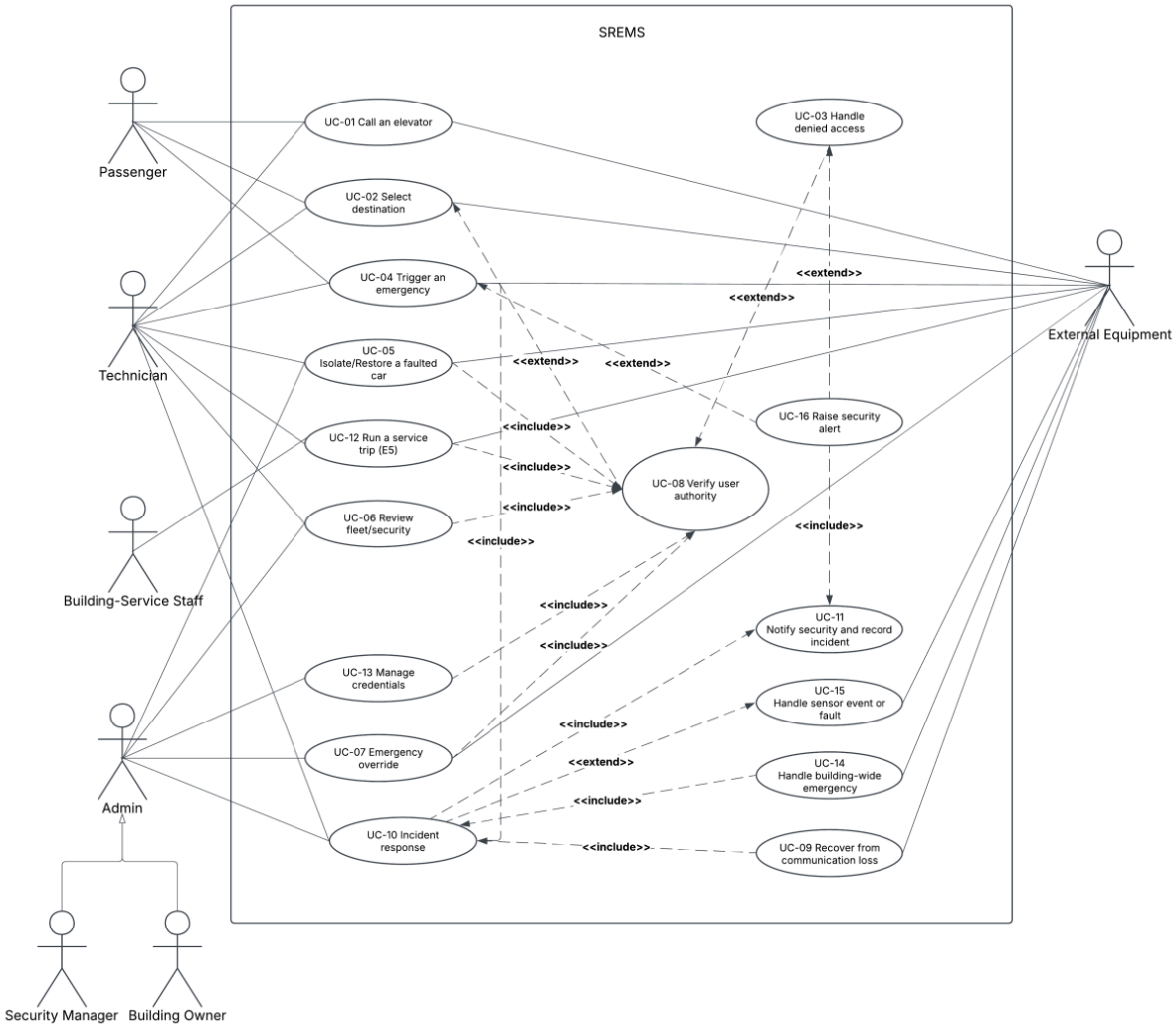


Figure 1: Use case diagram of SREMS.

2.3 User Characteristics and Stakeholder Requirements

The traceability chain is: stakeholder need -> measurable goal or constraint -> actor/use case -> system requirement -> test. The table below states what success means at the stakeholder level. The full chain from each need to its use cases, requirements, and tests is in Table 10. Requirements not listed there (FR-1.1, FR-1.12, FR-1.19, UI-1 to UI-3, UI-5, UI-8 to UI-10, SWI-1, COM-1 to COM-3, ANA-5.3, ANA-5.4, PERF-6, PERF-7, CAP-2, and the optional VT-12) are verified by inspection or demonstration.

Table 3: Stakeholder needs and measures

Stakeholder	Need / demand	Type	How to measure
Passenger	Short waiting time	Objective	Average and percentile wait time across named traffic profiles.
Passenger	Reach a permitted destination	Constraint	Valid requests complete; unauthorized destinations are rejected in the security test suite.
Passenger	Predictable door, load and status behavior	Constraint	No close command while an obstruction is active; overload is displayed and recorded when load data is available; no displayed state contradicts the latest valid state.
Passenger	Emergency help and accessible information	Constraint	Every valid test trigger creates a recorded incident and staff alert; floor/direction/arrival/emergency status is available as text and audio.
Building owner	Capacity and continuity	Objective	Queue length remains bounded under defined profiles; one car fault does not disable unrelated cars.
Building owner	Low downtime / maintenance burden	Objective	Downtime measured from recorded fault/maintenance intervals; diagnosis view exposes current state and recent requests.
Security manager	Restricted-floor access	Constraint	100% of unauthorized destination requests in the test suite are blocked before any command is issued.
Security manager	Awareness of suspicious/emergency activity	Constraint	Events are searchable; repeated failures and emergencies trigger alerts within the defined response target.
Security manager	Emergency override and rescue	Constraint	An authenticated Admin can dispatch an empty rescue car; the command is audited; the same command from other roles is rejected.
Technician	Safe maintenance access and fast diagnosis	Constraint / objective	Maintenance cars receive no new ordinary assignments; fault, state, and recent call history are available.
Building-service staff	Run service trips without interruption	Objective	E5 service runs (building-service staff or Technician) reach their destination without stopping for other calls; E5 shows OCCUPIED during a run.

Metric definitions are intentionally separated. Wait time measures service response to a hall call. Elevator travel time measures a recorded movement between floors. Passenger total journey time shall be reported only if the prototype can associate the original hall call and subsequent car call reliably; otherwise it shall not be presented as a per-passenger metric.

Actors at the system boundary

Table 4: Actors at the system boundary

Actor group	Actors	Responsibilities
Human actors	Passenger; Technician; Building-service staff; Admin (Security manager, Building owner)	Request service, use permitted functions, monitor status, respond to incidents, or perform maintenance according to role.
Operational inputs	Hall-call controls; car operating panel (including logical card-reader input); emergency/help trigger; technician service control	Provide input events to SREMS. These are logical interfaces, not a commitment to custom hardware.
External equipment	Elevator equipment (cars and door units); sensors (position, door, obstruction, load); building alarm system (fire and power status); or test stubs for any of them	Outside SREMS. Supply state and emergency events and accept supported commands through the elevator interface. Drawn as one secondary actor, External equipment, in Figure 1.

A card reader is treated as part of the car operating panel for this prototype unless the equipment requires a separate input. There is no requirement to build physical RFID, fingerprint, or camera hardware.

2.4 Constraints

- The building has floors B1 and 1-30. B1 is the parking level.
- Four residential elevators E1-E4 serve B1. Freight/service elevator E5 does not serve B1 in the baseline configuration.
- Hall calls require no passenger card. Travel to B1 requires no card. Residents must authenticate before selecting other destinations unless a configured policy explicitly grants access or a defined emergency policy applies.
- The default resident policy permits the assigned apartment floor and configured common floors. For this SRS, Floor 1 and B1 are common floors (B1 needs no card, SEC-2.5); rooftop access remains restricted unless explicitly authorized. Floor permissions must be configurable.
- E5 is for freight, goods, pets, waste transport, and building-service operations. Residents may also use E5 through the separate freight call, except to reach B1, but only to carry freight, goods, or pets. See FR-1.13.
- SREMS is software and shall not claim physical safety certification. It is tested through the elevator interface using test stubs.
- The team has six members and 14 weeks. Minimum demonstrable baseline: direction-aware dispatch, card/floor authorization and the B1 exception, repeated-failure alerting, one end-to-end emergency path, emergency override with a rescue car, fault/maintenance isolation, communication-loss recovery, tamper-evidence verification, repeatable wait-time reporting, the credential lifecycle (lost and cloned cards), and the medical emergency path. Conditional/stretch items are accessible floor/direction announcements (UI-6), native load/overload handling, additional emergency types, simulated public-address output, comparison against an ESRA built-in controller, and advanced analytics; implement these only if equipment support and time are confirmed, otherwise label them as test-stub input or defer them.
- The event and command set of the target elevator equipment or test stubs must be confirmed during implementation (OR-01).
- Priorities. Must: FR-1.1 to FR-1.13, FR-1.15, FR-1.16, FR-1.20 to FR-1.23, SEC-2.1 to SEC-2.19, EMR-3.1 to EMR-3.9, EMR-3.11, EMR-3.12, EMR-3.14, EMR-3.16, MNT-4.1 to MNT-4.9, MNT-4.11, ANA-5.1, ANA-5.2, ANA-5.8, COM-1 to COM-4, and PERF-1 to PERF-5. Should: FR-1.17 to FR-1.19, SEC-2.20 to SEC-2.22, EMR-3.10, EMR-3.13, EMR-3.15, MNT-4.10, ANA-5.3 to ANA-5.7, HW-1, HW-2, UI-1 to UI-10, PERF-6 to PERF-10, CAP-1, AVL-1, and DAT-1

to DAT-4. Could: FR-1.14 (overload), CAP-2, ANA-5.9, the optional benchmark (BEN-1 to BEN-5), and the simulated PA output in UI-7.

2.5 Assumptions and Dependencies

- Elevator equipment or test stubs can supply the events in Table 6 through the elevator interface.
- The exact communication mechanism of the elevator interface is not yet confirmed. The contract in Section 3.3 is logical and is mapped to the actual equipment or test stub during implementation (OR-01).
- If a needed event is not available natively, a test stub may provide it for demonstration, but reports must label it as test-stub input and not as native equipment behavior.
- Access cards and user identities can be represented digitally for the prototype. At least two Admins are available at all times or Admins must have backup cards, so that a lost Admin card can be suspended (SEC-2.17). If SREMS is down, the equipment continues in its own local mode (its own basic controller, Section 1.1) and no SREMS rule applies until it restarts (MNT-4.11).
- Emergency conditions, door obstruction, and load may be injected as repeatable test events if no native event exists.
- A persistent data store is available for request history, incidents, maintenance events, access policies, and audit records.
- The specific framework, database, and programming language remain implementation decisions. ESRA is needed only for the optional benchmark (Section 3.3.4).
- Requirements are refined after a minimum end-to-end prototype; every change is reviewed by all six team members and recorded in this SRS.
- Requirements qualified by 'where supported', 'where available', or 'when supplied' shall be tested against the data source recorded in OR-02. If the data is not native, the test uses a labelled test stub (SWI-2), so the requirement is still verified.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

SREMS shall minimize the number of distinct interface types. Role differences shall primarily be enforced through permissions and visible controls rather than separate, functionally duplicated applications.

UI-1 - Operational controls. The passenger-facing operational interface shall support hall-call input, destination selection, current floor/direction display, request status, authorization feedback, and emergency/help activation.

UI-2 - Technician interaction. Technicians shall use the same operational-control concept as passengers for normal car interaction. Technician-only maintenance actions shall become available only after a valid technician credential or equivalent simulated authorization. The prototype shall not require a separate passenger-specific and technician-specific button interface.

UI-3 - Monitoring dashboard. A monitoring dashboard shall be provided for admins (Security manager and Building owner). It shall show all elevator states, active requests, faults, emergencies, security alerts,

and relevant historical metrics. Role-based permissions shall determine which dashboard actions are available.

UI-4 - Technician diagnostic information. Fault and maintenance details required by technicians shall be available through the service/diagnostic view or a role-restricted dashboard panel; a separate standalone technician dashboard is not required.

UI-5 - Clear state display. Floor, direction, door status, and out-of-service/emergency state shall be shown with text and/or icons. Critical states shall not be conveyed by color alone.

UI-6 - Accessibility baseline. The prototype shall provide screen-reader-compatible labels and a non-visual announcement for floor/direction changes, destination arrival, and emergency status. If browser/OS speech is unavailable in the chosen environment, equivalent accessible text/status information shall remain available.

UI-7 - Resident notification. An emergency shall produce an occupant-facing message on the affected elevator/passenger interface and an alert on the monitoring dashboard. A physical public-address system is out of scope; a simulated building-wide PA notification may be added only if time and the demo environment support it.

UI-8 - Car operating panel. The panel inside each car shall provide destination keys (B1 and 1 to 30), a card reader, door-open and door-close keys, an emergency/help key with a separate medical option, a floor and direction display, a door and load status indicator, and an area for authorization feedback.

UI-9 - Hall panel. Each floor from 1 to 30 shall provide a separate freight key for E5 (B1 has none, FR-1.13) and an up key and a down key (one key at the top floor and at the lowest served floor) and a display of the assigned car, its direction, and its arrival, with a non-visual arrival announcement (UI-6). A hall panel has no card reader and no cancel control (FR-1.20).

UI-10 - Dashboard views. The monitoring dashboard shall provide these views, with role permissions applied (SEC-2.13): fleet status for E1 to E5, active requests, incidents and alerts, audit and incident search, analytics, and the audit-integrity check. The technician diagnostic view (UI-4) shall provide maintenance mode, return to service, and the diagnosis data of MNT-4.5.

3.1.2 Hardware Interfaces

No custom physical hardware is required for the academic prototype. Hall-call buttons, car operating panels, card readers, door units, load sensors, displays, emergency triggers, and technician service controls are logical inputs/outputs represented by software or elevator-interface events. If an actual device is used for demonstration, its protocol and limitations shall be documented separately.

HW-1 - Sensor inputs. SREMS shall treat these sensors as external equipment that reports through the elevator interface: floor position, door state, door obstruction, load, fire alarm input and power status input. The in-car emergency/help trigger is an operational input (Table 4), not a sensor. Each sensor event shall carry the sensor type, the elevator identifier when it is car-specific, the value, and a timestamp, and shall be validated as in SEC-2.10.

HW-2 - Sensor fault handling. If a sensor reports an illegal value, stops reporting for longer than the stale threshold (5 seconds by default), or reports a value that contradicts the latest valid state, SREMS shall mark that sensor FAULTY, record a fault, notify staff, and apply the safest defined behavior. An obstruction sensor stuck active keeps the close command inhibited and escalates under FR-1.15. A load sensor fault disables overload handling for that car without removing the car from dispatch. A position sensor fault makes the car state stale (FR-1.17).

3.1.3 External Software Interfaces

No external software is mandatory. The only external software identified is ESRA, which may be connected for the optional benchmark (Section 3.3.4). The frontend, application services, and database are internal design components and are not treated as external software interfaces in this SRS. No external authentication provider is required; identity/card credentials may be simulated within SREMS.

SWI-1 - Elevator interface mapping. Before full implementation, the team shall document the equipment or test stub used, its launch procedure, the API or extension point used, supported input/state events, supported control operations, timing model, and any unsupported requirements.

SWI-2 - Test-stub labelling. If a required event or command is not provided natively, the implementation shall use a clearly labelled test-stub input. It shall not represent a stub-generated event as coming from real equipment.

3.1.4 Communications Interfaces

COM-1 - Interface boundary. All communication between SREMS and the elevator equipment or test stubs shall pass through one elevator interface that validates messages and maps them to stable SREMS logical events and commands. No other code path shall reach the controller.

COM-2 - Message identification. Each request/command/event shall carry, where supported, a unique identifier, event/command type, elevator identifier, timestamp, and payload. Invalid or duplicate messages shall be handled deterministically.

COM-3 - Acknowledgement and timeout. The elevator interface shall distinguish an acknowledged command, a rejected command, a timed-out command, and a command whose execution status is unknown.

COM-4 - No unsafe blind retry. After a command times out or communication is lost, SREMS shall not blindly resend an uncertain command. It shall refresh/reconcile elevator state and request status before retrying, to avoid duplicate control actions.

3.2 Functional Requirements

3.2.1 Mode 1 - Monitoring and Passenger Service

FR-1.1 - Elevator registration. The system shall maintain unique identities E1-E5 and distinguish the four residential elevators from the freight/service elevator.

FR-1.2 - Current state monitoring. The system shall display the latest valid floor, direction, movement state, door state, availability, and fault/emergency status for each value the elevator interface supplies.

FR-1.3 - Hall call. A passenger shall be able to submit an ordinary up/down hall call (served by E1 to E4) or a freight call (served by E5) without presenting a card. The system shall record the origin floor, call type, requested direction, request identifier, and accepted time.

FR-1.4 - Direction-aware dispatch. For a normal residential hall call, the system shall select an eligible residential elevator using predicted service time. The decision shall consider the requesting direction, current position, direction of travel, existing commitments, and availability; it shall not select by absolute floor distance alone. A freight call is assigned to E5 only (FR-1.13).

FR-1.5 - Dispatch tie-break. If two eligible cars have the same predicted service time within the implementation precision, the system shall use a deterministic tie-breaker: select the lower elevator identifier (E1 before E2, then E3, then E4).

FR-1.6 - Dispatch exclusion. A car in emergency, maintenance, blocking fault, stale/untrusted state, OCCUPIED by a service run (FR-1.13), or otherwise unavailable status shall not receive a new ordinary passenger assignment.

FR-1.7 - Hall-call lifecycle. Each hall request shall follow the defined states RECEIVED -> QUEUED -> ASSIGNED -> APPROACHING -> ARRIVED -> COMPLETED. A request may instead enter CANCELLED (set only by the system or an Admin, FR-1.20) or FAILED with a recorded reason. During an emergency, a request may be BLOCKED/REQUIRES_RECONCILIATION (EMR-3.3) and returns to QUEUED once the state is reconciled. If the assigned car becomes ineligible, the request also returns from ASSIGNED or APPROACHING to QUEUED and is reassigned (FR-1.18, FR-1.23). COMPLETED means the car arrived at the requesting floor; it does not imply the passenger has completed the whole journey.

FR-1.8 - Car-call lifecycle. Each car call shall be recorded as RECEIVED -> AUTHORIZATION_CHECK -> QUEUED -> IN_TRANSIT -> COMPLETED. A denied request shall enter REJECTED; a request affected by an emergency shall enter BLOCKED/REQUIRES_RECONCILIATION (EMR-3.3); a cancelled/failed request shall include a reason. These states shall not be conflated with hall-call completion.

FR-1.9 - Destination selection. A passenger shall be able to submit a destination request from inside a car. For restricted destinations, the authorization decision shall be completed before a destination command is sent through the elevator interface.

FR-1.10 - B1 access. A request to travel to B1 shall be accepted without an access card, provided a residential elevator is eligible and B1 service is not disabled by a defined emergency or maintenance policy. E5 shall not be assigned to B1.

FR-1.11 - Travel progress. During travel, the interface shall display the latest valid floor and direction and, where supported by the equipment, progress toward the destination.

FR-1.12 - Arrival and timestamps. When the elevator interface reports arrival at a destination, the system shall record the arrival timestamp and update the corresponding car-call state. If no interface timestamp is available, SREMS shall record its local receipt time and label the metric accordingly.

FR-1.13 - Freight/service elevator. E5 shall not be assigned to B1. Residents and staff may request E5 through the freight call (FR-1.3), under the same access rules (SEC-2.3); SREMS cannot detect what a rider carries, so the purpose is a building rule shown on E5's signage and in the resident rules. E5 trips stay in the audit log (ANA-5.1), so an Admin can review misuse. It shall also support authorized freight, goods, pets, waste transport, and building-service use cases. A service run, once started, shall go directly to its destination without stopping for other calls. A service run may be started only by authorized building-service staff or a Technician. While such a run is active, E5 shall show OCCUPIED, and shall accept no new assignment or freight call. A freight call made while E5 is OCCUPIED or otherwise ineligible (FR-1.6) is rejected with a message on the hall panel, recorded, and not queued.

FR-1.14 - Load/overload status. When load data is available, the system shall display load status and record an overload event when the load exceeds the threshold defined in CAP-2. If no load data is available, this function shall be labelled as test-stub input or deferred.

FR-1.15 - Door obstruction. While a valid obstruction event is active, the system shall inhibit/reject a normal door-close command and record the obstruction. If the obstruction persists beyond a configurable threshold (initial prototype default: 10 seconds), the system shall create a warning/incident and notify security. It shall never bypass the obstruction by forcing a close command. The door state values are CLOSED, OPEN, CLOSING, and OBSTRUCTED; OBSTRUCTED lasts from the obstruction event until it clears.

FR-1.16 - Door/floor consistency. The system shall not issue a normal door-open command unless the latest valid state indicates that the car is at a valid floor. This is a software rule and not a claim of physical interlock certification.

FR-1.17 - State consistency. The system shall not present an older state as current after a newer valid state has been received. If current state cannot be confirmed, the interface shall visibly mark it stale or unavailable.

FR-1.18 - Wait-time measurement. The system shall measure wait time as the elapsed time between acceptance of a hall call and arrival of the assigned elevator at the call floor. If a request is reassigned, the record shall retain the original acceptance time and record the reassignment event.

FR-1.19 - Travel-time measurement. Where valid departure and arrival timestamps exist, the system shall measure elevator travel time between those events. It shall not claim this value equals an individual passenger ride time unless the corresponding passenger hall and car calls can be reliably linked.

FR-1.20 - Call cancellation. A passenger shall not be able to cancel a submitted hall call or car call. Only the system, for a recorded failure or reassignment reason, or an Admin may set a request to CANCELLED, and the actor and reason shall be recorded.

FR-1.21 - Empty hall call. If the assigned car opens its doors at the origin floor and no car call, load change, or door-obstruction event follows within 30 seconds (default, configurable), the request shall still be COMPLETED and shall be flagged NO_BOARDING. A NO_BOARDING request stays in the wait-time statistics, because wait time ends when the car arrives (FR-1.18), and is counted per floor. Where the equipment supplies no boarding evidence, a labelled test stub provides it (SWI-2).

FR-1.22 - Duplicate and repeated calls. A hall call for a floor and direction that already has an active request shall be merged into that request and shall not create a second assignment. If five or more requests from the same floor are flagged NO_BOARDING within 10 minutes (default, configurable), SREMS shall raise a low-severity security alert that links the underlying events. SREMS shall keep serving hall calls during an alert and shall not block them (SEC-2.4).

FR-1.23 - Group operation. E1 to E4 shall operate as one dispatch group for ordinary hall calls, and E5 serves freight calls (FR-1.13): one dispatcher assigns every hall call of either type, a hall call has at most one assigned car at a time, and the call is reassigned to another eligible car if its car becomes ineligible.

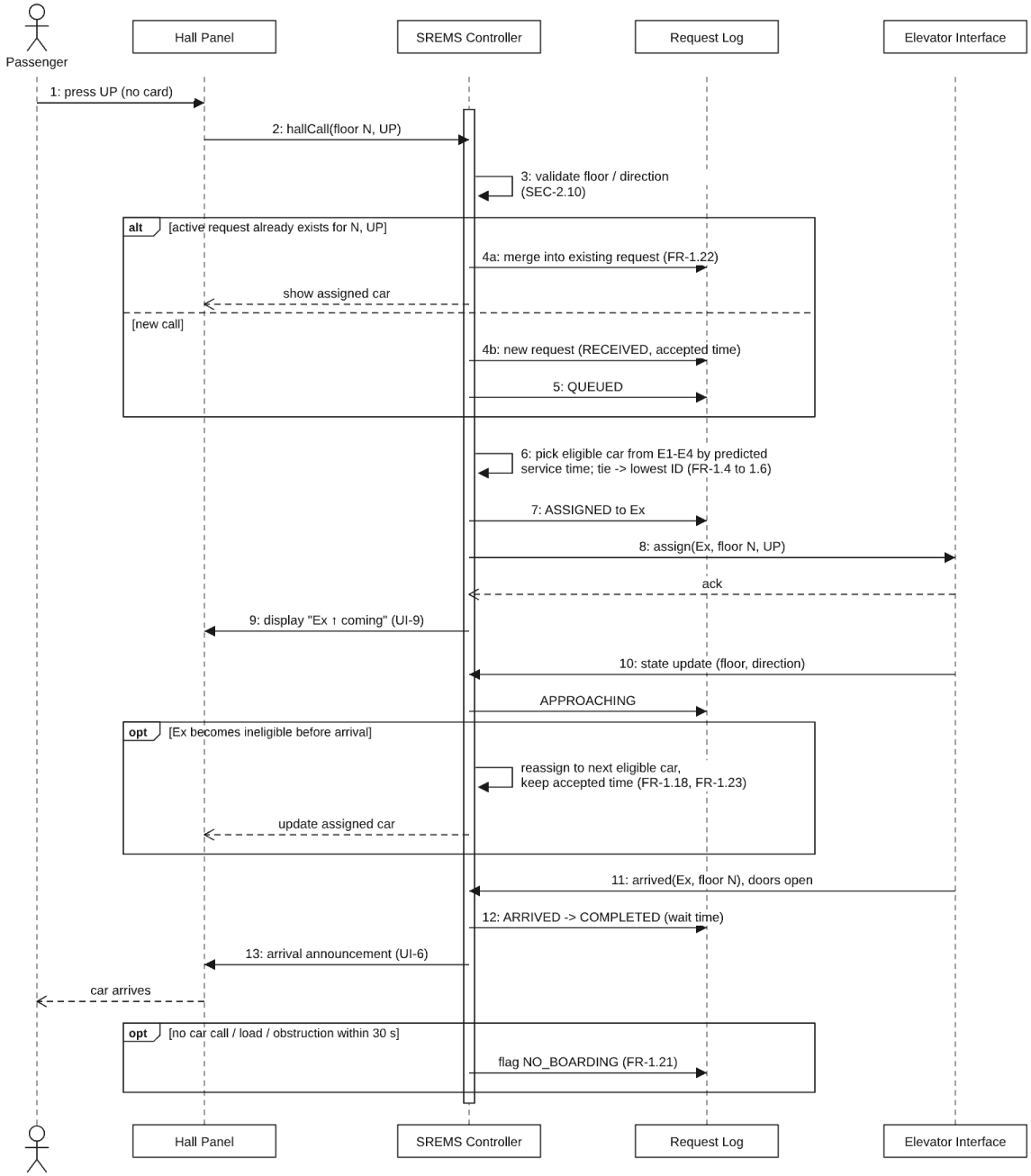


Figure 2: Hall call and dispatch sequence diagram (UC-01; FR-1.3 to FR-1.7, FR-1.21 to FR-1.23).

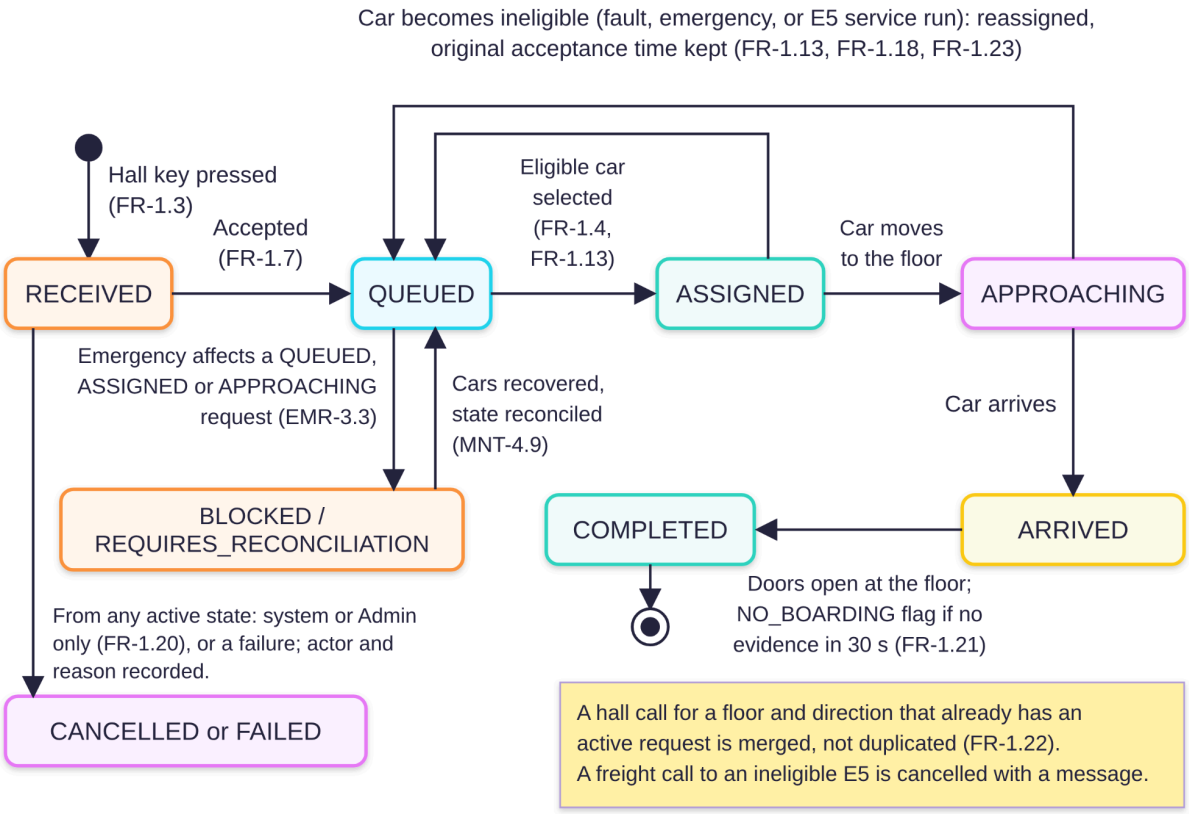


Figure 3: State machine of a hall call (RECEIVED, QUEUED, ASSIGNED, APPROACHING, ARRIVED, COMPLETED, CANCELLED or FAILED, BLOCKED, with ASSIGNED or APPROACHING back to QUEUED on reassignment; FR-1.7, FR-1.20 to FR-1.22).

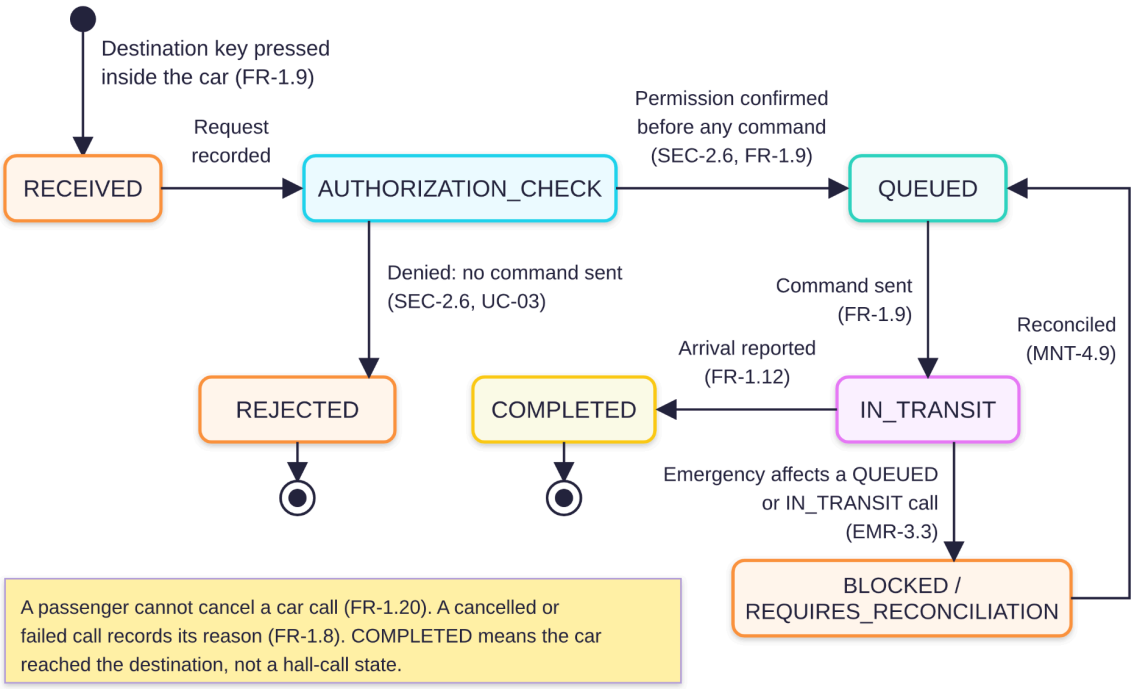


Figure 4: State machine of a car call (RECEIVED, AUTHORIZATION_CHECK, QUEUED, IN_TRANSIT, COMPLETED, REJECTED, BLOCKED; FR-1.8, FR-1.9, FR-1.20).

3.2.2 Mode 2 - Access Control and Security

SEC-2.1 - Simulated card identity. The system shall represent an access card with a unique credential identifier linked to a user identity and permission set. The prototype shall use simulated card data; fingerprint or camera authentication is not part of the core scope.

SEC-2.2 - Role-based authorization. The system shall enforce separate permissions for passengers, technicians, building-service staff, and Admins (Security manager and Building owner). Building-service staff may start E5 service runs (FR-1.13) and hold no maintenance or Admin rights.

SEC-2.3 - Resident floor policy. By default, a resident may select the apartment floor assigned to the resident and configured common floors. For this SRS, Floor 1 and B1 are common floors; B1 needs no card (SEC-2.5). Rooftop access and all other floors shall remain restricted unless explicitly granted.

SEC-2.4 - Hall-call exception. Hall-call submission shall remain available without passenger authentication.

SEC-2.5 - B1 exception. A request for B1 shall be accepted without a card, subject to residential elevator availability and defined emergency/maintenance restrictions.

SEC-2.6 - Authorization denial. An unauthorized destination command shall be rejected before it reaches the elevator interface. The passenger shall receive clear feedback and the denial shall be recorded with timestamp, card/user identifier where available, elevator, destination, and reason.

SEC-2.7 - Repeated failure detection. Every failed authentication/authorization decision shall be recorded, but a single denial shall not automatically create a high-priority alert. The system shall generate an alert when the number of failed attempts reaches a configurable threshold. Initial prototype default: three failed attempts for the same credential within 60 seconds.

SEC-2.8 - Security dashboard alerts. Security alerts shall include severity, timestamp, relevant user/card identifier where available, elevator, attempted action, result, and reason. Alerts shall link to the underlying events when possible.

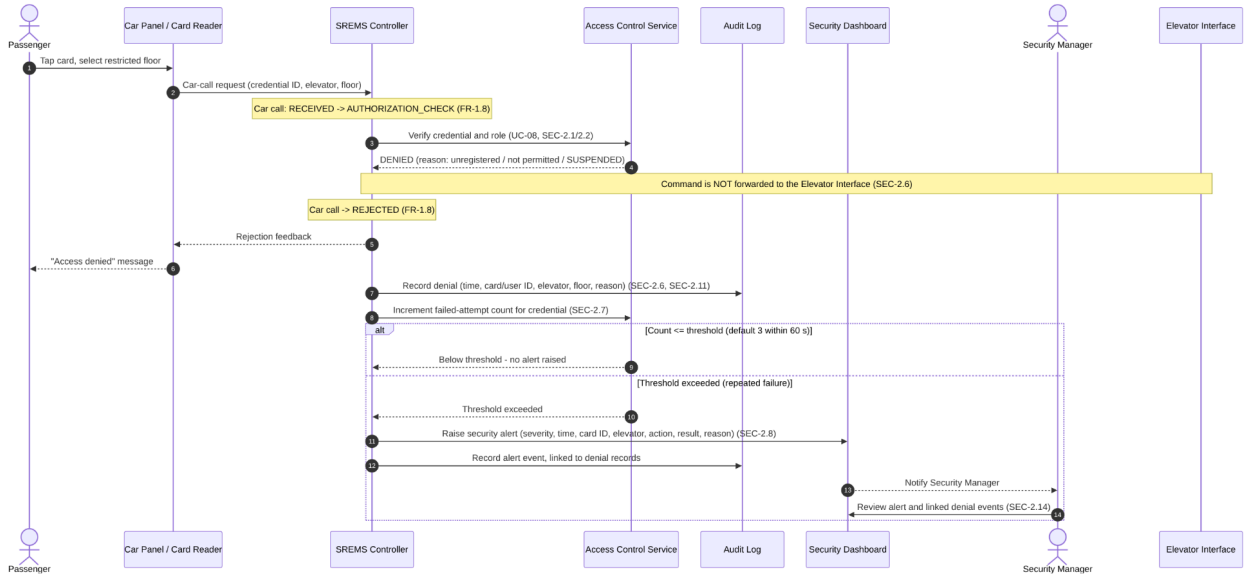


Figure 5: Denied access and repeated failure sequence diagram (UC-03, UC-08; SEC-2.6, SEC-2.7, SEC-2.8).

SEC-2.9 - Control-command permissions. Commands that change elevator availability, maintenance state, emergency state, or access policy shall require an authorized role and shall be audited.

SEC-2.10 - External input validation. All incoming elevator messages shall be validated for recognized message type, known elevator, valid floor range (B1-30), legal state value, and permissible state transition before acceptance.

SEC-2.11 - Security audit events. The system shall record security-sensitive actions including successful/failed authentication, destination denial, permission changes, maintenance isolation, emergency commands, manual overrides, and rejected external messages.

SEC-2.12 - Audit log protection. Audit records shall be protected against unauthorized modification or deletion after creation. Any unauthorized modification or removal shall be detectable through an integrity-verification mechanism. The specific implementation mechanism is a design decision.

SEC-2.13 - Least privilege. Users shall not invoke operations outside their assigned role and explicit permissions, regardless of which interface path is used.

SEC-2.14 - Security investigation. Admins shall be able to search and review access denials, repeated failures, emergency events, and relevant audit records by time range, user/card, elevator, event type, and status where data is available.

SEC-2.15 - Authentication method. Authentication shall use a credential (card identity) only: presenting the card at the car operating panel or the service control for passengers, technicians, and building-service staff, and signing in with the same credential at the monitoring dashboard for Admins. The prototype shall not require a second factor. A dashboard session shall end after 15 minutes without activity (configurable).

SEC-2.16 - Credential lifecycle. A credential shall be in one of the states ACTIVE, SUSPENDED, REVOKED, or EXPIRED. An Admin shall be able to issue, replace, suspend, reinstate, and revoke a credential, and each action shall be audited (SEC-2.11). A suspension records its reason (LOST or

CLONE_SUSPECTED, SEC-2.17, SEC-2.18) and time. A SUSPENDED credential may be reinstated or revoked. A REVOKED credential is final and is never reinstated; a new credential is issued instead. A credential that is not ACTIVE shall be handled as an unauthorized credential for restricted destinations (SEC-2.6). It shall not affect hall calls or travel to B1 (SEC-2.4, SEC-2.5).

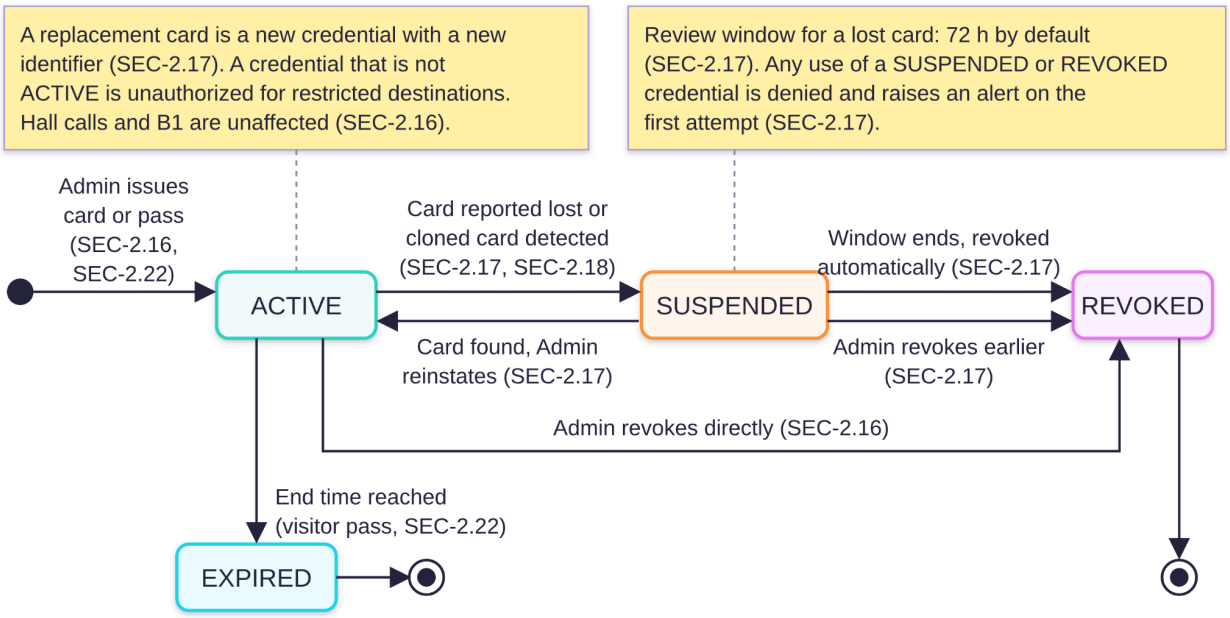


Figure 6: state machine of a credential (ACTIVE, SUSPENDED, REVOKED, EXPIRED; SEC-2.15 to SEC-2.18, SEC-2.22).

SEC-2.17 - Lost card. When a card holder reports a card lost, an Admin shall first suspend it (SUSPENDED, reason LOST) for a review window of 72 hours by default (configurable, 48 to 72 hours), because a lost card is often found again. For a lost Admin card, another Admin shall do so. An Admin shall revoke the card if it is not found, or earlier if misuse is suspected, and issue a replacement with a new credential identifier. Issuing a replacement revokes the lost card. The replacement shall receive the permission set of the lost card only after the Admin confirms the identity of the card holder. Any use of a SUSPENDED (lost) or REVOKED credential shall be denied, recorded, and shall raise a security alert on the first attempt, without waiting for the SEC-2.7 threshold. If the card is found within the window, an Admin may reinstate it after confirming the identity of the card holder. When the window ends, the card is revoked automatically.

SEC-2.18 - Duplicated or cloned card. If one credential identifier is presented in two different cars within 10 seconds (default, configurable), SREMS shall deny the later request, raise a security alert that names both cars, and set the credential to SUSPENDED until an Admin reviews it.

SEC-2.19 - Unregistered credential. A credential identifier that is not registered shall be denied for restricted destinations and counted as a failed attempt under SEC-2.7, keyed by that identifier. If more than 10 distinct unregistered identifiers are presented within 5 minutes (default, configurable), SREMS shall also raise one security alert for suspected card scanning.

SEC-2.20 - Command ordering. Control commands that act on the same car shall be applied in the order SREMS receives them. A command that conflicts with the state left by an earlier command, for example a rescue dispatch for a car that is in maintenance, shall be rejected with a reason and audited (SEC-2.9).

SEC-2.21 - One presentation, one destination. A credential presentation shall authorize exactly one restricted destination selection. A passenger without a credential may ride to a floor that an authorized passenger in the same car has selected, but cannot select any other restricted floor. Riding along with another passenger is not detected and is not treated as a violation.

SEC-2.22 - Visitor pass. At a resident's request, an Admin shall be able to issue a visitor pass: a credential limited to named floors that expires after a set time (default 4 hours). An expired pass follows SEC-2.16, and every issue and expiry shall be audited (SEC-2.11).

3.2.3 Mode 3 - Emergency and Safety

For this SRS, an “emergency state” is a software state with defined consequences. It does not mean that SREMS can guarantee physical safety or independently determine real-world evacuation behavior.

EMR-3.1 - Emergency trigger. A passenger or technician shall be able to trigger an emergency/help event, including a medical emergency, from inside a car. The trigger shall never require a card, role, or other permission check. The in-car control shall offer two options, HELP and MEDICAL. A MEDICAL trigger starts EMR-3.14. A HELP trigger creates an incident, a dashboard alert, and an in-car message, but it does not by itself take the car out of service. The car becomes EMERGENCY/UNAVAILABLE as a stuck car only when the equipment reports a stuck event or an Admin confirms the HELP request on the dashboard (Table 5).

EMR-3.2 - Emergency-state behavior. When a valid car-specific emergency is received (a MEDICAL trigger, a stuck event, or an Admin-confirmed HELP request), SREMS shall mark the car EMERGENCY/UNAVAILABLE, exclude it from new ordinary assignments, show the state, notify security, create an incident, and preserve affected request records. Only explicitly authorized emergency commands, and the responder-floor run of EMR-3.14, may be issued while this state is active.

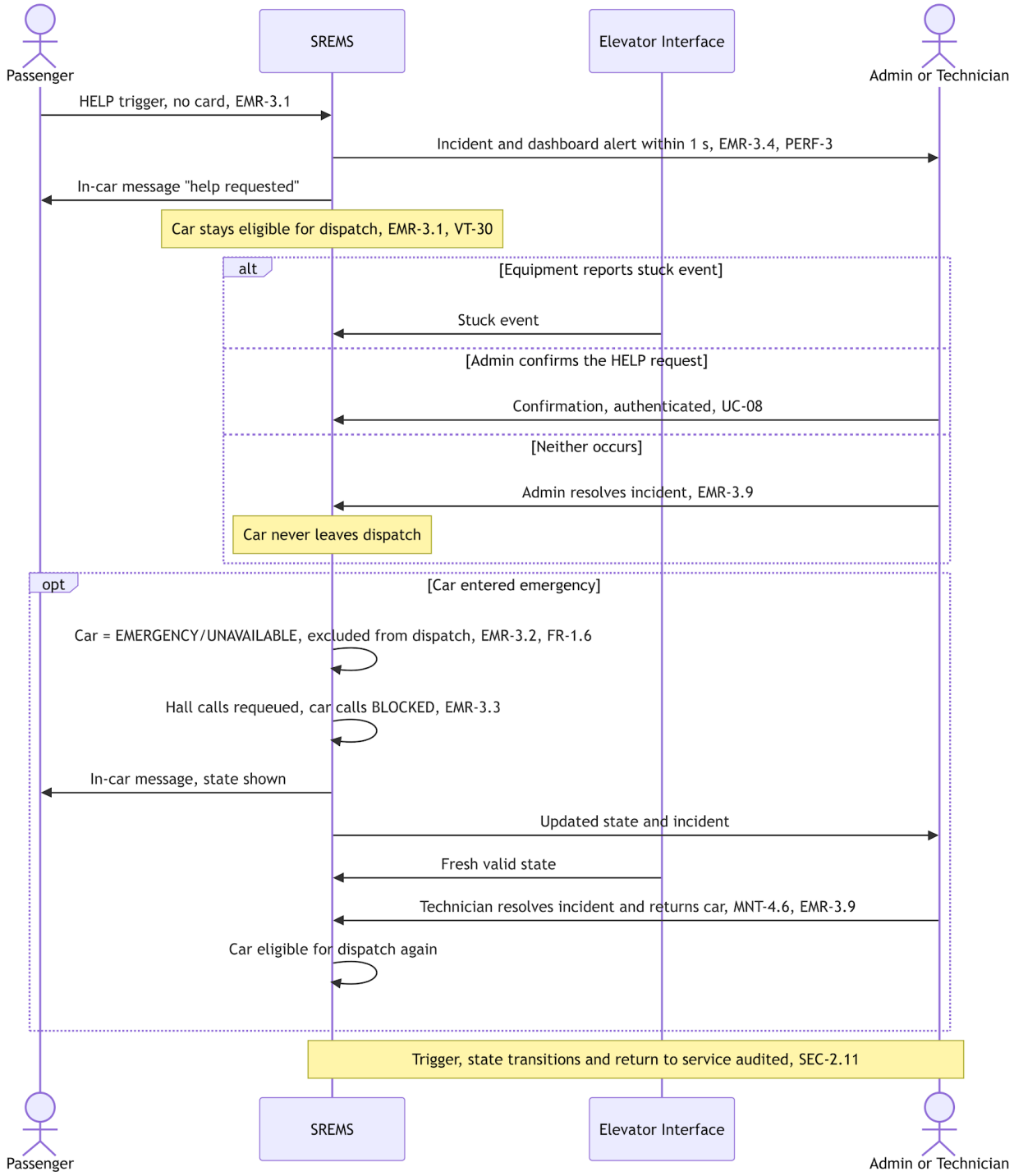


Figure 7: Emergency sequence for a HELP trigger and a stuck car; ‘Elevator Interface’ denotes the elevator equipment or a labelled test stub (SWI-2)

EMR-3.3 - Active requests during emergencies. Requests assigned to an affected car shall be marked BLOCKED/REQUIRES_RECONCILIATION or requeued to another eligible car according to the scenario

policy. The system shall not silently drop them or mark them complete without evidence from the elevator interface.

EMR-3.4 - Emergency event record. Each emergency incident shall contain the event type, elevator/building scope, current valid floor and direction where known, timestamp, trigger/source, state transition, notification status, related requests, and resolution history.

EMR-3.5 - Emergency modes. The prototype implements fire, power failure, medical emergency, stuck car, door obstruction, communication failure, and overload, as mapped in Table 5. Overload handling is conditional on OR-09 and shall be implemented only if a reliable load-data source or test-stub mechanism is confirmed. Each mode shall have a documented trigger, state change, request handling, notification, and recovery condition. Door obstruction and overload leave the car in its current availability state, and communication failure marks state STALE/UNTRUSTED (MNT-4.8); fire, power failure, medical emergency, and stuck car place cars in EMERGENCY/UNAVAILABLE (EMR-3.2).

EMR-3.6 - Building-wide events. For fire or building-wide power failure, the system shall apply the configured building-emergency policy to affected cars and suspend ordinary assignments. The default building-emergency policy is: all affected cars enter EMERGENCY/UNAVAILABLE, no new ordinary assignment is made, active requests are handled as in EMR-3.3, and no recall or evacuation movement is commanded. Any further behavior, such as recalling cars to a floor, shall be added to this policy as an explicit, tested rule. It shall not invent or claim a real-world evacuation/recall behavior that has not been defined and tested.

EMR-3.7 - Emergency notification. On receipt of a valid emergency, the system shall record the incident and surface an alert on the monitoring dashboard within the time in PERF-3. The affected-car interface shall show an occupant-facing message and, where supported by the prototype, an audible announcement. A public-address broadcast is optional and shall be clearly simulated if no PA interface exists.

EMR-3.8 - Emergency override. Only an Admin (Security manager or Building owner) may issue emergency overrides, and only those explicitly permitted by the configured policy. Each override shall be authenticated (UC-08), validated, audited, and associated with a reason.

EMR-3.9 - Emergency resolution. Authorized staff shall be able to acknowledge, handle, and resolve an incident. An emergency car shall not return to normal dispatch until the required recovery condition is satisfied and an authorized state transition is recorded. An incident not resolved within a configurable time threshold shall be escalated by re-notifying security, and the escalation shall be recorded.

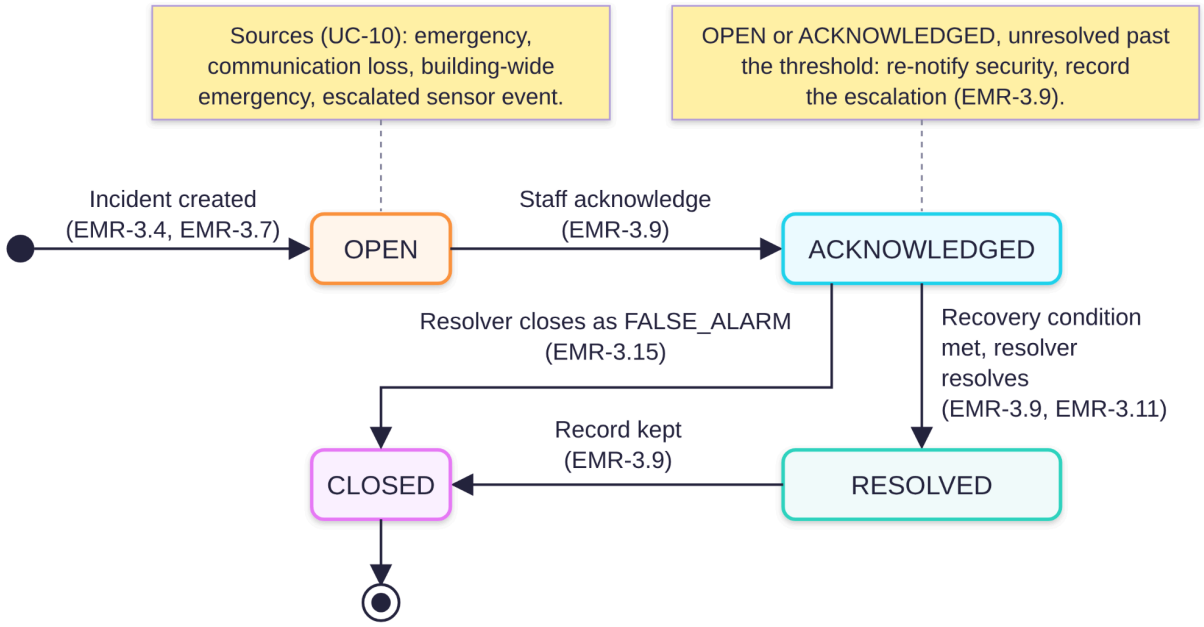


Figure 8: state machine of an incident (OPEN, ACKNOWLEDGED, RESOLVED, CLOSED including FALSE_ALARM; escalation after the threshold; EMR-3.9, EMR-3.15).

EMR-3.10 - Door safety. Door-open and door-close restrictions are specified in FR-1.15 and FR-1.16. They shall be verified by test cases and shall not be represented as hardware-certified protection.

EMR-3.11 - Incident-response mapping. For each emergency mode implemented in the prototype, SREMS shall apply the trigger, scope, state change, request handling, notification, and recovery condition defined in Table 5. A mode that is not listed is not implemented. Recovery shall be performed only by the resolver role named for that mode (an Admin may always act in place of a Technician; a Technician may not act in place of an Admin) and shall be audited (EMR-3.9).

EMR-3.12 - Rescue car. When a car is in an emergency state with occupants, or a medical emergency is reported (EMR-3.16), an Admin may dispatch a rescue car to the floor where the affected car is stopped or the ill person is waiting (UC-07). The rescue car shall be an eligible car that serves that floor, selected by the FR-1.4 rule; the affected car and E5 shall never be selected. SREMS shall requeue the rescue car's pending ordinary assignments to other eligible cars (EMR-3.3) and shall dispatch it only when it carries no passengers. If load or occupancy data is unavailable, the Admin shall confirm that the car is empty (OR-15). The dispatch shall be audited, and the affected car's emergency state and incident remain open until resolved (EMR-3.9).

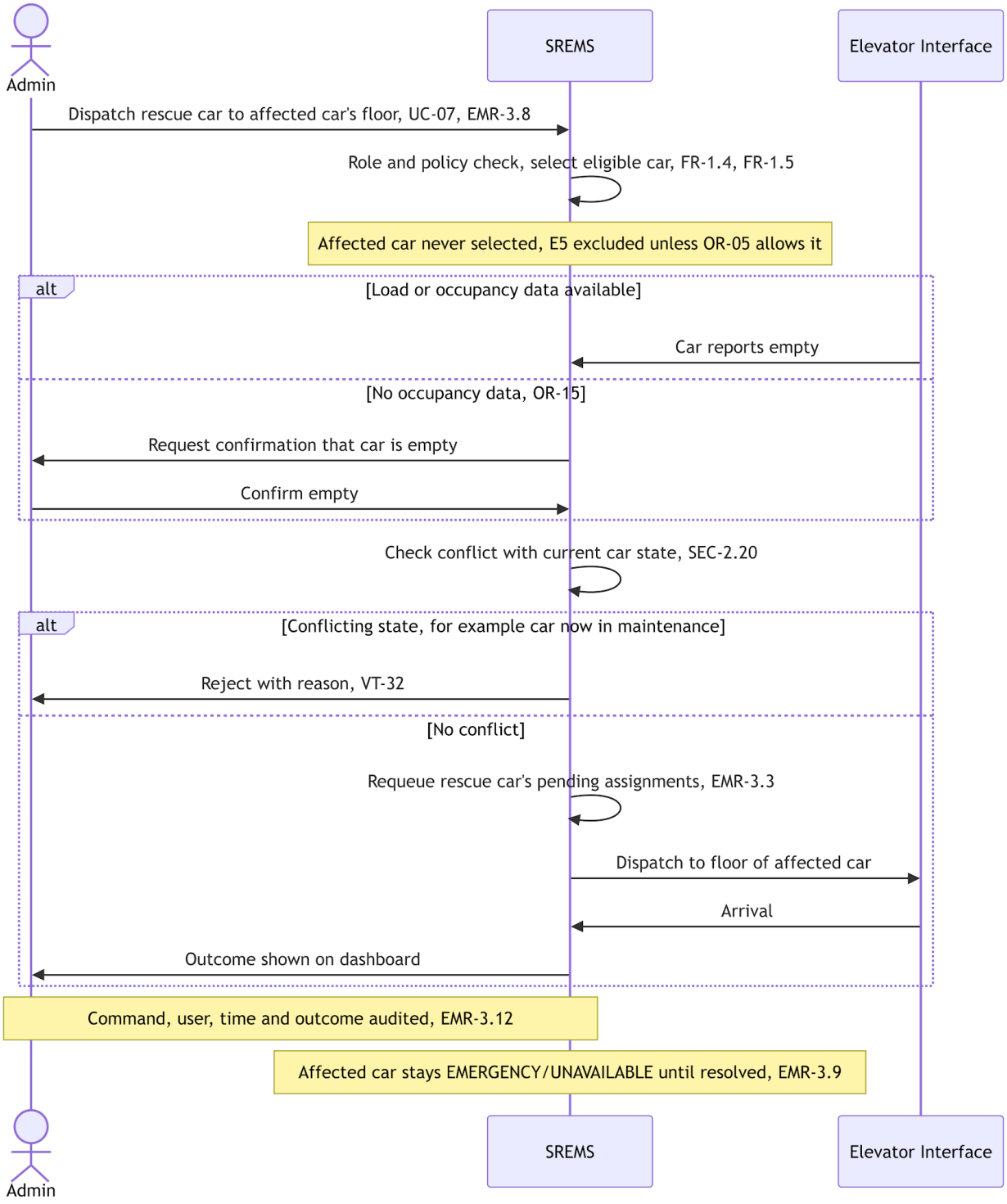


Figure 9: Rescue car dispatch sequence; 'Elevator Interface' denotes the elevator equipment or a labelled test stub (SWI-2).

EMR-3.13 - Concurrent incidents. If several emergency modes are active, the most restrictive one applies, in this order: fire, power failure, communication failure, medical emergency, stuck car, door obstruction, overload. Lower-ranked incidents stay open and notified; only their dispatch restrictions are superseded.

EMR-3.14 - Medical emergency. A passenger shall be able to trigger a medical emergency from the in-car emergency control without presenting a card or any other permission. SREMS shall treat it as a single-car emergency with the highest incident severity: the car is marked EMERGENCY/UNAVAILABLE, security is notified immediately with the car, floor, and direction, and the occupant interface shows that help has been requested. Hall calls assigned to the car are requeued, and its car calls are marked BLOCKED/REQUIRES_RECONCILIATION (EMR-3.3). Unless the car is stuck or a higher-ranked emergency applies (EMR-3.13), it shall then travel non-stop to the configured responder floor (default Floor 1) and open its doors there (FR-1.16). A stuck car follows the stuck car response in Table 5. An Admin resolves the incident (EMR-3.9). When the ill person has not yet entered a car, or the Admin is informed of the emergency by telephone, the Admin may send a rescue car (EMR-3.16). If the affected car is E5 or a rescue car, the same behavior applies, and any rescue dispatch to that car is cancelled by the system and reported to the Admin.

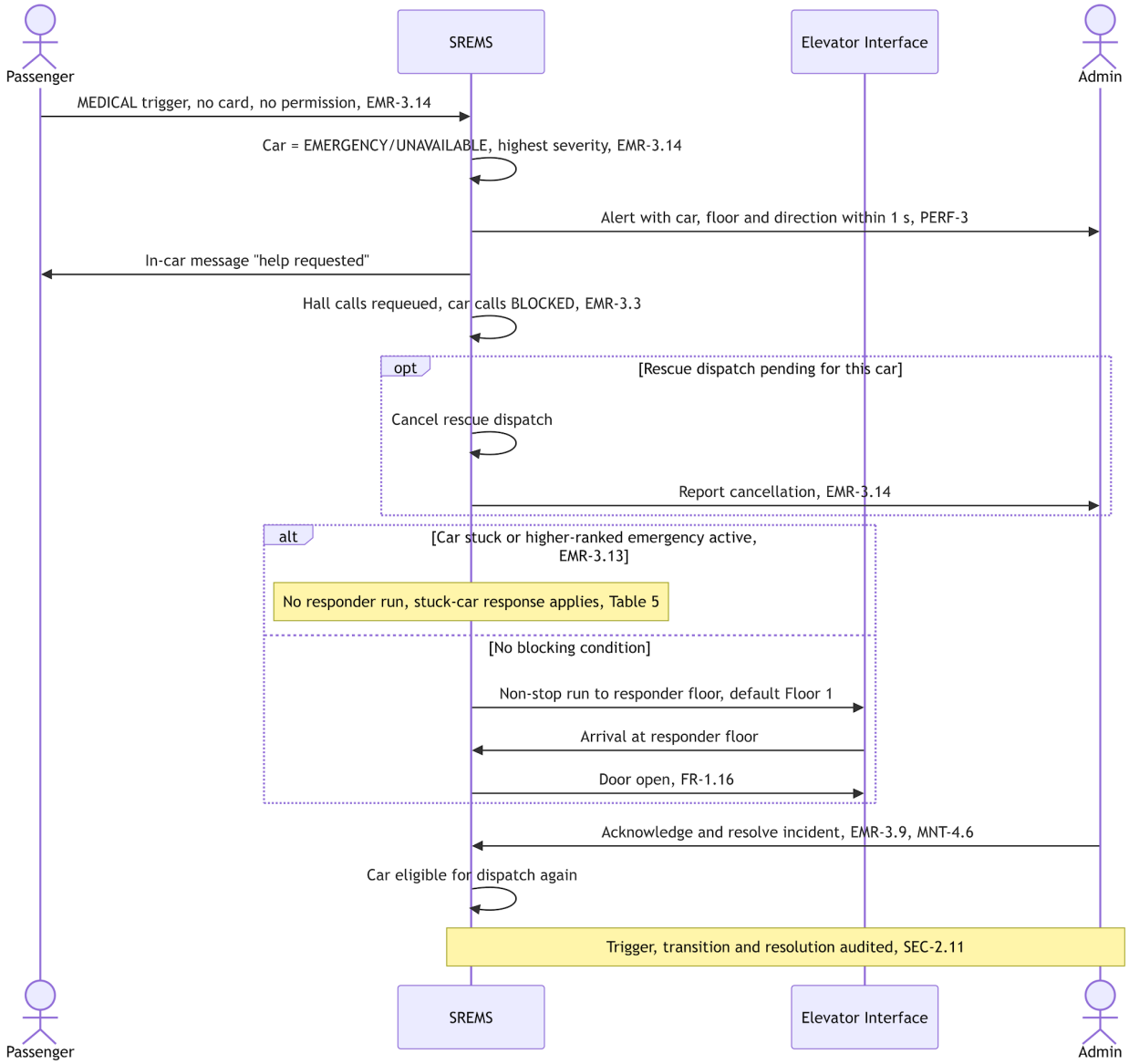


Figure 10: Medical emergency sequence; ‘Elevator Interface’ denotes the elevator equipment or a

labelled test stub (SWI-2).

EMR-3.15 - Repeated emergency triggers. Every emergency trigger shall be recorded with its car and time and shall be treated as valid, including repeated triggers. If one car receives more than three triggers within 60 seconds (default, configurable), SREMS shall add one low-severity security alert that links them. Staff may close an incident as FALSE_ALARM when they resolve it (EMR-3.9). This also applies to misuse of the medical option, which an Admin can review in the incident record (EMR-3.4); in-car video is out of scope (Section 2.3).

EMR-3.16 - Medical rescue car. An Admin informed of a medical emergency by telephone shall be able to record it on the dashboard as an incident (EMR-3.4), and may send a rescue car (EMR-3.12) to that floor either before or after the ill person has entered a car. The Admin sets how long the rescue car waits with its doors open (default 120 s, maximum 600 s); this wait is not a door fault. When the Admin confirms that the ill person is aboard, that car is treated as the affected car of EMR-3.14 and runs non-stop to the responder floor. If the Admin cancels or the wait expires, the car is released to normal dispatch and the incident stays open. The dispatch, wait time, confirmation, and release shall be audited.

Table 5: Emergency types and system responses

Emergency type	Trigger	Scope	State change	Request handling	Notification	Recovery (resolver)
Fire	Fire alarm event	Building-wide	All cars: EMERGENCY/UNAVAILABLE	No new assignments. Active requests: BLOCKED/REQUIRES_RECONCILIATION. No recall or evacuation (EMR-3.6).	Monitoring dashboard alert; occupant message in car	Policy cleared and incident acknowledged. Admin resolves.
Power failure	No-power event	Building-wide	All cars: EMERGENCY/UNAVAILABLE	No new assignments. Active requests: BLOCKED/REQUIRES_RECONCILIATION.	Monitoring dashboard alert; occupant message in car	Power restored, fresh state snapshot, reconciliation done (MNT-4.9). Technician returns cars to service.
Stuck car	Stuck event from the equipment, or Admin confirmation of an in-car HELP request	Single car	Car: EMERGENCY/UNAVAILABLE	Hall calls requeued to another eligible car; car calls: BLOCKED.	Monitoring dashboard alert; occupant message in affected car	Recovery checks passed (MNT-4.6). Technician resolves.
Door obstruction	Obstruction event	Single car	Door state OBSTRUCTED (FR-1.15)	Close command inhibited, never forced. Requests stay queued. After the threshold (default 10 s) a warning/incident is created (FR-1.15).	Security alert after threshold	Obstruction cleared. Admin closes the incident.
Communication failure	Connection lost or timed out	All cars	State marked STALE/UNTRUSTED	No new control commands. In-flight commands: UNKNOWN/PENDING_RECONCILIATION, no blind retry (MNT-4.8).	Staff notified	Connection restored, fresh snapshot, reconciliation (MNT-4.9). Technician resolves.

Emergency type	Trigger	Scope	State change	Request handling	Notification	Recovery (resolver)
Medical emergency	In-car medical trigger (no card or permission needed), or a telephone report recorded by an Admin (EMR-3.16)	Single car	Car: EMERGENCY/UNAVAILABLE, highest incident severity	Hall calls requeued to another eligible car; car calls: BLOCKED/REQUIRES_RECONCILIATION. The car travels non-stop to the responder floor (default Floor 1) and opens its doors there, unless it is stuck or a higher-ranked emergency applies (EMR-3.13, EMR-3.14). An Admin may also send a rescue car and set how long it waits (EMR-3.16).	Immediate dashboard alert with car, floor, and direction; in-car message that help was requested	Incident acknowledged and recovery checks passed (MNT-4.6). Admin resolves.
Overload (conditional, OR-09)	Load above threshold (test-stub input if no load data)	Single car	Overload event recorded	Display and record only; dispatch unchanged (FR-1.14).	In-car message; dashboard event	Load below threshold. Admin closes the incident.

3.2.4 Mode 4 - Fault and Maintenance Management

MNT-4.1 - Fault reception. The system shall receive or detect fault events and mark the affected elevator with the relevant fault state.

MNT-4.2 - Fault visibility. Active faults and recent state/call history shall be accessible to authorized staff. The monitoring dashboard shall show fault state and incident severity; technicians shall be able to inspect diagnostic details.

MNT-4.3 - Maintenance mode. An authenticated technician or admin shall be able to place an elevator into maintenance mode using an authorized service control.

MNT-4.4 - Assignment blocking. An elevator in maintenance mode is excluded from ordinary dispatch as specified in FR-1.6.

MNT-4.5 - Diagnosis information. For the selected elevator, the diagnostic view shall provide current/last-valid state, active fault, state freshness, and recent call history sufficient to reproduce the issue where records exist.

MNT-4.6 - Return to service. Only an authenticated technician or admin shall return a car to service after the recovery checks pass. The recovery checks are: the car reports a fresh valid state, no fault is active for it, its doors are not obstructed, and the person returning it to service confirms the result. The action shall be audited.

MNT-4.7 - Maintenance record. The system shall record elevator, technician, fault/cause where known, maintenance action, start/end time, result, and return-to-service state.

MNT-4.8 - Communication-loss response. When the connection to the elevator equipment or test stub is lost or times out, SREMS shall mark the affected state as stale, notify staff, stop sending new control

commands, and preserve requests without falsely completing them. An in-flight command with unknown outcome shall be marked UNKNOWN/PENDING_RECONCILIATION and shall not be blindly retried.

MNT-4.9 - Reconnection and reconciliation. After connection recovery, SREMS shall obtain a fresh valid state snapshot, reconcile outstanding request/command identifiers, resolve unknown outcomes, and resume control only after state consistency is restored. Duplicate execution shall be avoided where the interface supports identifiers/idempotency.

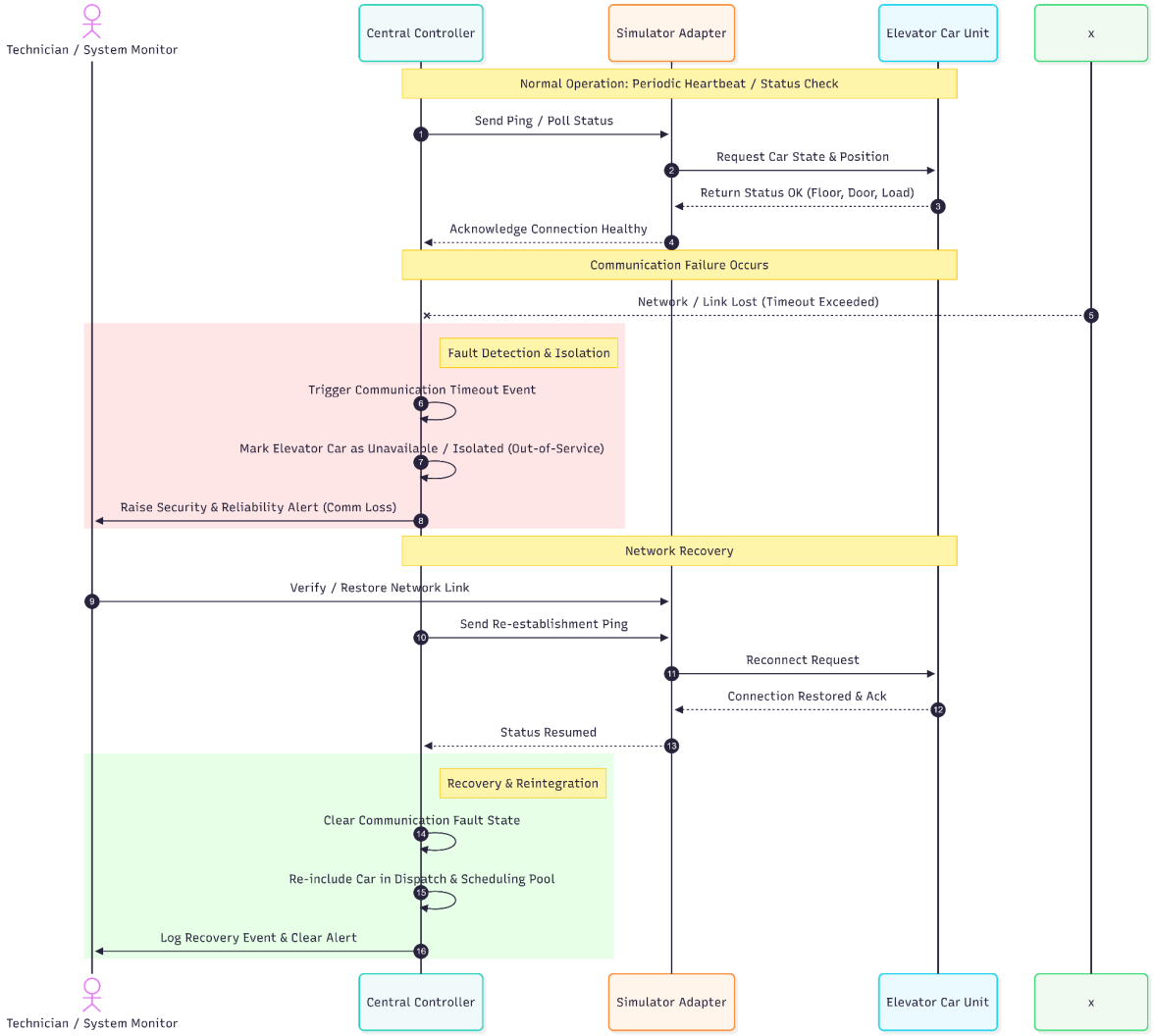


Figure 11: Sequence diagram for communication loss and recovery

MNT-4.10 - Door obstruction persistence. Escalation of a persistent obstruction to security, and the prohibition on forcing the door closed, are specified in FR-1.15.

MNT-4.11 - SREMS Failure. If SREMS itself is down, it issues no commands and the equipment continues in its own local mode (Section 2.5). If SREMS is running but its data store is unavailable, restricted destinations shall fail closed (be denied), except B1 and destinations opened by an active emergency policy. A security decision that cannot be written to the audit log shall be rejected.

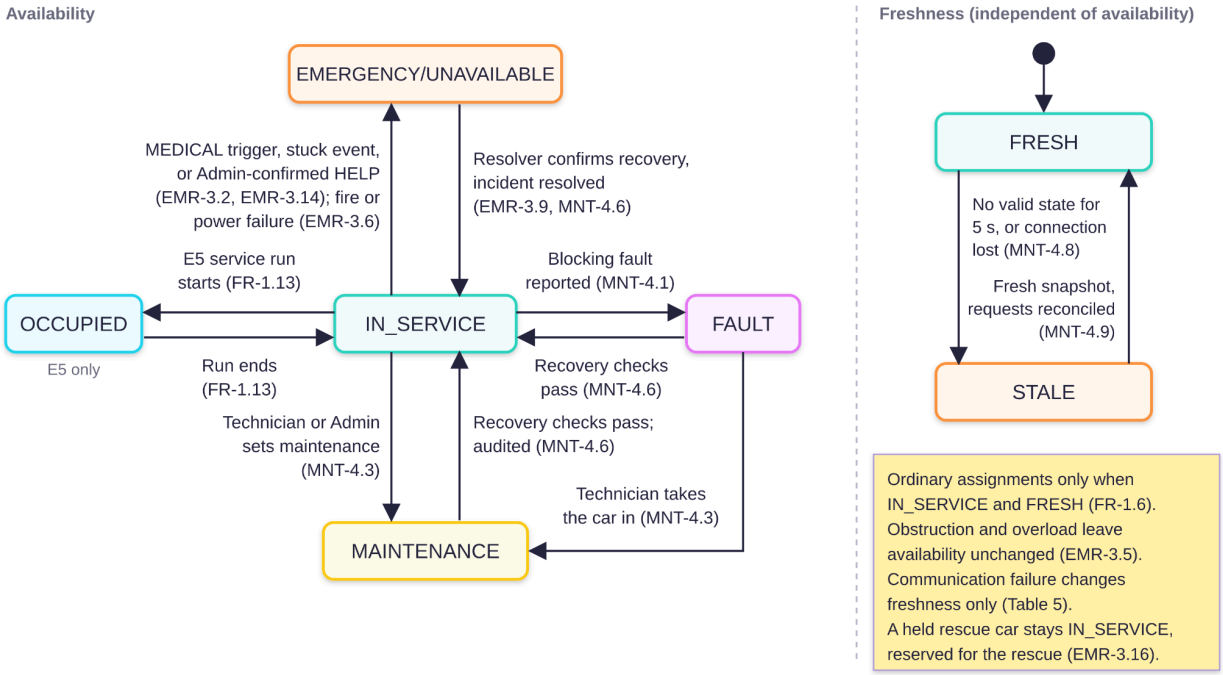


Figure 12: state machine of a car (IN_SERVICE, OCCUPIED for E5, MAINTENANCE, FAULT, EMERGENCY/UNAVAILABLE, STALE; MNT-4.1 to MNT-4.9, EMR-3.2).

3.2.5 Mode 5 - Analytics and Audit Management

ANA-5.1 - Historical event persistence. The system shall persist hall calls, car calls, assignments, trips, fault transitions, emergencies, communication-loss events, maintenance operations, authorization decisions, and security alerts.

ANA-5.2 - Wait-time metrics. The system shall report average, median, and percentile wait time by selected period and traffic profile when the underlying timestamp data is available.

ANA-5.3 - Travel-time metrics. The system shall report elevator travel time between recorded departures and arrivals when available. It shall identify missing/estimated timestamps and shall not relabel whole-car operating time as passenger ride time.

ANA-5.4 - Utilization metrics. The system shall report per-car trip count, starts, travelled distance or a supported equivalent, and time in operational/fault/maintenance states when those data can be obtained or derived consistently.

ANA-5.5 - Downtime. The system shall calculate each car's recorded downtime from fault/maintenance start and end events. Intervals still open at report time shall be identified as ongoing.

ANA-5.6 - Security metrics. The system shall summarize authorization denials, failed authentication attempts, repeated-failure alerts, external validation failures, and emergency incidents.

ANA-5.7 - Incident search. Authorized staff shall search incident and audit records by elevator, user/card, event type, status, and time range where those fields exist.

ANA-5.8 - Audit integrity verification. An Admin shall be able to run audit-integrity verification and view whether the recorded chain is intact, including the first detected invalid record when one is found.

ANA-5.9 - Report export. The system shall export per-request timestamps/outcomes and aggregate performance/reliability/security measures for a recorded run, including optional benchmark runs (Section 3.3.4).

3.3 Elevator Interface and Optional Benchmarking

Sections 3.3.1 to 3.3.3 define the logical contract between SREMS and the elevator equipment or test stubs. Exact message formats are design decisions that follow the interface confirmation in OR-01. Section 3.3.4 describes the optional benchmark.

3.3.1 State and Input Events

Table 6: Elevator interface events

Logical event	Minimum useful data	Required handling
Elevator state	Elevator ID; floor/position; direction; movement/availability; timestamp; door/fault/emergency state when supported.	Validate ID, range, state value, timestamp, and legal transition; update the latest valid state.
Hall call	Request ID; origin floor; up/down direction; acceptance time.	Record request, validate floor/direction, and submit to dispatch.
Car call	Request ID; elevator ID; destination; user/card context when available; time.	Authorize before issuing the destination command; record allow/deny and result.
Door/obstruction	Elevator ID; door state; obstruction flag; event time.	Inhibit close while obstructed; record duration; escalate after threshold.
Load/overload	Elevator ID; load/capacity or threshold; event time.	Display and record only if supplied by the equipment or a labelled test stub.
Emergency/fault	Type; scope; elevator ID if car-specific; time; known state/location.	Apply the documented mode policy, create an incident, notify staff, and preserve related requests.
Communication	Connect/disconnect; timeout; invalid message; reconnect/snapshot status.	Mark state stale, stop new commands while disconnected, alert staff, then reconcile before resuming.

3.3.2 Control Commands

- Assign an eligible elevator to an accepted hall call, where the elevator interface supports external assignment/control.
- Submit or schedule a car destination after authorization has succeeded.
- Issue door commands only if the equipment or test stub exposes such commands and the project state policy permits the operation.
- Set or represent maintenance/service availability through the equipment or through SREMS state if no explicit maintenance command is supported.
- Issue supported emergency and override commands, including rescue-car dispatch, only after role authorization and policy validation.
- Request a state refresh or reconnect snapshot, where supported.

For every logical command, the interface documentation shall state the input fields, return/acknowledgement behavior, expected state event, error/timeout behavior, and whether the operation is supported by the equipment or provided by a test stub.

3.3.3 Validation, Recovery, and Fault Isolation

- Reject floor values outside B1 through 30 and unknown elevator IDs.
- Reject unknown message/command types and invalid state values.
- Reject state transitions that conflict with the project state model; retain the last valid state for display and mark it stale when freshness is uncertain.
- Reject role-protected commands from unauthorized users and audit the rejection.
- On connection loss, stop issuing new control commands and avoid automatic replay of commands with unknown execution status.
- On reconnection, request a current state snapshot and reconcile active request IDs before normal control resumes.
- Log invalid messages and communication faults with enough context for repeatable diagnosis.

3.3.4 Optional Benchmark Runs

Benchmark runs are optional. No other requirement in this SRS depends on them or on ESRA, except that the peak and off-peak traffic profiles of OR-10 are also used by PERF-9, PERF-10, and VT-23, so they shall be defined even if no benchmark is run. The requirements below apply only if the team decides to run a benchmark (OR-14).

BEN-1 - Scenario configuration. If a benchmark is run, the team shall define named off-peak, peak, and stress traffic profiles, including the request-generation assumptions needed to reproduce each run.

BEN-2 - Repeatability. The same scenario configuration shall be replayable with recorded settings and comparable outputs. Random inputs, if used, shall use a recorded seed where the tool allows it.

BEN-3 - Recorded results. Each benchmark shall record request arrival, assigned car, car-arrival time, destination completion where available, reassignments, dropped/failed requests, fault/maintenance intervals, and relevant security events.

BEN-4 - Aggregate reports. Each benchmark report shall include average/median/percentile wait time, request queue size, per-car service/utilization measures, faults/downtime, and security-event counts where the data is available.

BEN-5 - Baseline comparison. If ESRA exposes a built-in controller through a documented interface, the team may compare SREMS dispatch against that baseline using identical traffic profiles. If no compatible baseline interface exists, the report shall present repeatable SREMS results without claiming a direct comparison.

3.4 Performance and Non-Functional Requirements

PERF-1 - User feedback. For ordinary interface actions excluding car travel time, the system shall provide visible feedback within 1 second under the defined normal test conditions.

PERF-2 - State refresh. A valid state received through the elevator interface shall be reflected in the monitoring display within 1 second under normal test conditions.

PERF-3 - Critical alert latency. A critical emergency/security event shall be visible on the security dashboard within 1 second of SREMS receiving the event under normal test conditions.

PERF-4 - Five-car operation. The system shall handle state updates and concurrent request activity for all five elevators in the agreed test traffic profiles.

PERF-5 - Queue boundedness. The request queue shall remain bounded under each defined load profile. If requests exceed the backlog limit (default 100 active requests, configurable), the system shall expose the backlog and shall not silently discard requests.

PERF-6 - Record persistence. All critical events generated during the defined healthy-database test scenarios shall be present in the event history after the scenario completes.

PERF-7 - Threshold review. Performance targets shall be re-evaluated after the first end-to-end prototype establishes realistic update rates. Any changed target shall be recorded with a reason in this SRS.

PERF-8 - Traffic profiles. Wait-time targets are verified under a named peak profile and a named off-peak profile (OR-10). The default peak windows are 07:00 to 09:00 and 17:00 to 20:00, and the default off-peak window is 22:00 to 05:00. SREMS shall use the same dispatch rule (FR-1.4) at all times and shall not change its behavior by time of day.

PERF-9 - Peak targets. Under the peak traffic profile (OR-10), the average wait time shall not exceed 60 seconds and the 95th percentile shall not exceed 120 seconds, with at least E1 to E4 in the dispatch group (FR-1.23).

PERF-10 - Off-peak target. Under the off-peak traffic profile (OR-10), the average wait time shall not exceed 90 seconds.

CAP-1 - Capacity. SREMS shall support at least 600 apartments (30 floors with 20 apartments each), 3,000 registered credentials (about five per apartment), and 100 simultaneous active requests across E1 to E5 without breaching PERF-1 to PERF-3.

CAP-2 - Car load rating. The rated load is 1,000 kg (13 persons) for each residential car (E1 to E4) and 1,600 kg for the freight/pet car (E5). A car is overloaded when the reported load exceeds 110% of its rated load (configurable). SREMS uses these values only to display load status and to record the overload event (FR-1.14); it does not measure or control the load. The ratings are configuration parameters, and the load source is open (OR-09).

AVL-1 - Availability. Monitoring and dispatch shall be available at least 99.5% of each calendar month (about 3.6 hours of downtime in a 30-day month), excluding announced maintenance windows. Downtime of SREMS itself shall be measured from its restart log and from a heartbeat check every 10 seconds. Car downtime is reported under ANA-5.5.

DAT-1 - Stored data. SREMS shall store requests, assignments, trips, incidents, maintenance records, credentials and permissions, access policies, configuration, and audit records in the persistent data store (Section 2.5).

DAT-2 - Retention. Request, incident, and maintenance records shall be kept for at least 12 months. Audit records shall be kept for at least 12 months and shall be archived rather than deleted, so that the integrity check of ANA-5.8 still covers the whole chain.

DAT-3 - Backup and recovery. The data store shall be backed up at least once every 24 hours, and a restore shall take at most 1 hour, so at most 24 hours of non-audit records can be lost. Audit records shall also be copied to the backup location as they are written, so a restore loses none of them. Backup copies shall be kept apart from the live data store under the same access restrictions, and the audit chain shall be verified (ANA-5.8) after each restore.

DAT-4 - Data protection. Stored data shall be reachable only through role permission checks (SEC-2.13). A credential record shall hold only the credential identifier, the linked user identity, and the permission set, and no biometric data. Data shall be encrypted in transit and at rest where the chosen platform supports it.

The metric, target, verification method, and test for the measurable requirements in this section are listed in Table 11.

3.5 Design Constraints

- The project should use open-source or freely available tools where practical.
- Core dispatch, authorization, incident management, and audit logic shall be testable without physical elevator equipment, using test stubs.
- All elevator traffic shall pass through the elevator interface; alternate code paths shall not bypass validation, authorization, or logging.
- Floor access policy, card permissions, elevator type/capacity, request thresholds, and obstruction thresholds shall be configuration data where practical.
- The prototype shall use simulated card credentials; biometric recognition and physical card-reader hardware are not core requirements.
- Tamper-evidence implementation is a design decision; it must satisfy the modification-detection test in SEC-2.12 and ANA-5.8.
- The system shall not claim physical safety, PA delivery, or equipment capabilities that the project cannot demonstrate.
- The SRS, prototype, and tests shall be updated together as requirements are refined.

3.6 Software System Attributes

Table 7: Software system attributes

Attribute	Requirement and verification focus
Security	Enforce role and floor permissions; reject unauthorized commands before they reach the elevator interface; alert on repeated failures; validate external messages; provide auditable and tamper-evident records.
Reliability	Maintain explicit request/state transitions; exclude unavailable cars; preserve critical event records; reconcile state after communication recovery; do not blindly replay commands.
Availability	A fault affecting one car shall not disable monitoring/control of unrelated eligible cars, unless the event is explicitly defined as building-wide. The availability target is AVL-1.
Data integrity	Validate identifiers, floors, state transitions, request IDs, and audit records; detect duplicate/invalid external messages and integrity-check audit history.
Usability and accessibility	Use minimal operational controls, clear status labels, non-color status indicators, screen-reader-compatible text, and non-visual announcements for key elevator state changes.
Maintainability	Keep equipment-specific code inside the elevator interface; log failures with diagnostic context; expose current/last-valid state and recent call history to authorized technicians.
Scalability/configurability	The building floor set, elevator identities/types, capacities, common floors, and access permissions shall be configurable rather than embedded in dispatch logic.
Auditability	Security-sensitive events shall be traceable to time, action, outcome, reason, and user/card/elevator identifiers when available.

4 Open Requirements and Requirements Validation Plan

The following questions remain intentionally open because they depend on the target elevator equipment or building-specific policy. The team shall close them through inspection, prototyping, and review during implementation and record each closure as a change to this SRS.

4.1 Open Requirements

Table 8: Open requirements

ID	Open topic	Resolution required
OR-01	Elevator interface mechanism	Confirm the target equipment or test stub, its API or extension points, launch sequence, state refresh method, and whether external dispatch commands are supported.
OR-02	Available interface data	Record which floor, direction, door, load, timing, fault, and emergency fields are native equipment data, derivable values, or test-stub events.
OR-03	Command semantics	For each supported command, identify acknowledgement, completion event, timeout, and recovery behavior.
OR-04	Building access policy	Confirm which common floors are accessible to all residents. Proposed baseline: assigned apartment floor plus Floor 1; rooftop and other floors require explicit permission; B1 remains the no-card exception.
OR-05	Freight/service policy	Confirm which waste/freight/service request types E5 handles, the OCCUPIED rule, and whether a service run is non-stop (FR-1.13).
OR-06	Emergency-mode mapping	Closed for fire, power failure, medical emergency, stuck car, door obstruction, communication failure, and overload (see EMR-3.11). Earthquake, emergency stop, and elevator malfunction are not implemented in the prototype unless a mapping is added. The item is retained here for traceability and needs no further action.
OR-07	Failed-authentication threshold	Confirm or retain the prototype default of three failed attempts for the same credential within 60 seconds.
OR-08	Obstruction threshold	Confirm or retain the prototype default of 10 seconds before alert escalation for continuous door obstruction.
OR-09	Load/capacity source	Confirm whether the equipment supplies load, and confirm the rated loads of CAP-2 (proposed: 1,000 kg for E1 to E4 and 1,600 kg for E5). If not, decide whether a labelled test stub is feasible or the feature is deferred.
OR-10	Benchmark traffic profiles	Define request distributions (including the arrival rate per floor and the share of up, down, and B1 trips) and run duration for off-peak, peak, and stress tests, so that PERF-9, PERF-10, and CAP-1 can be verified; record random seeds where supported.
OR-11	Tamper-evident implementation	Select a mechanism such as hash chaining and define how integrity verification reports the first mismatch.
OR-12	Accessibility implementation	Confirm the speech/announcement method supported by the chosen UI framework and test screen-reader-compatible labels.
OR-13	Emergency resident notice	Confirm whether a simulated building-wide/public-address message is feasible; the baseline remains the affected-car interface plus security dashboard.
OR-14	Benchmark baseline	Confirm whether a built-in ESRA controller can be compared fairly to SREMS using the same traffic profile. Do not claim comparative results until verified.

ID	Open topic	Resolution required
OR-15	Rescue-car occupancy	Confirm how SREMS determines that a rescue car is empty (equipment load data or Admin confirmation). Proposed default: Admin confirmation. E5 is never a rescue car (EMR-3.12).
OR-16	Responder floor and medical response	Confirm how an Admin confirms an in-car HELP request (EMR-3.1) and the responder floor (proposed: Floor 1), whether the doors stay open there, and whether a non-stop run with other passengers aboard is acceptable (EMR-3.14). It also covers how a telephone report reaches the Admin (proposed: outside SREMS, then entered by the Admin in the dashboard), the default and maximum rescue wait time (proposed: 120 s and 600 s), and what happens if a different person boards the held car (proposed: the Admin does not confirm and cancels the hold).
OR-17	Call abuse thresholds	Confirm or retain the defaults: 30-second empty-call window, five NO_BOARDING requests from one floor in 10 minutes, and more than three emergency triggers on one car in 60 seconds (FR-1.21, FR-1.22, EMR-3.15).
OR-18	Traffic schedule and targets	Confirm the peak and off-peak profile windows, the 60-second and 90-second average wait targets, and the capacity and availability targets (PERF-8 to PERF-10, CAP-1, AVL-1).
OR-19	Credential security parameters	Confirm the 10-second duplicate-use window, the card-scanning alert (10 identifiers in 5 minutes), the 4-hour visitor pass, the 72-hour lost-card review window (48 to 72 hours), the 15-minute dashboard session timeout, and how an Admin confirms the identity of a resident who replaces a card (SEC-2.15 to SEC-2.18).
OR-20	Data retention and backup	Confirm 12-month retention, the daily backup, the 1-hour restore target, and the backup location (DAT-2, DAT-3).

4.2 Validation and Prototyping

During implementation, the team shall complete a small end-to-end prototype and verify the requirements below. A test passes only when the observable outcomes specified in the criterion all occur; a screen mock-up alone does not count as validation.

Table 9: Validation and prototype tests

Test ID	Scenario	Pass criteria
VT-01	Hall call and dispatch	Without a card, an up/down hall call is recorded. An eligible elevator is chosen using direction/position/queue state. The assignment and timestamps are recorded.
VT-02	Unauthorized destination	A card without permission requests a restricted floor. No destination command is sent through the elevator interface; denial feedback appears; a denial audit event is stored. A user without the required role also cannot read audit or credential data by any interface path (SEC-2.13, DAT-4).
VT-03	B1 exception	A passenger submits a destination request for B1 without a card. An eligible residential car may serve it; E5 is never assigned to B1.
VT-04	Repeated access failures	Three failed attempts for the same credential within 60 seconds (prototype default) create a security alert; individual attempts are recorded; the restricted commands remain blocked.
VT-05	Direction-aware dispatch	Use a known scenario where the physically closest car is moving away or has conflicting stops. Verify that dispatch evaluates direction/commitments and applies the deterministic tie-break when scores tie.

Test ID	Scenario	Pass criteria
VT-06	Persistent door obstruction	With obstruction active, no close command is issued. When obstruction exceeds the configured threshold, the event is recorded and the dashboard displays a security/operations warning.
VT-07	Emergency path	A valid emergency trigger changes the affected car/building state according to the configured policy, blocks ordinary assignments as defined, displays occupant-facing status, creates an incident, and surfaces a security alert within 1 second of event receipt under normal test conditions.
VT-08	Communication loss	During a request, cut the interface connection. State is marked stale; no new commands are sent; requests are preserved; an in-flight unknown command is not blindly retried. After reconnecting, a fresh snapshot is obtained and the request state is reconciled before control resumes.
VT-09	Maintenance isolation	Put an elevator in maintenance, and in a second run inject a blocking fault in one car. New ordinary assignments exclude it; diagnosis information remains available; only an authenticated technician or admin can return it to service; each transition is logged. In both runs, the other cars keep receiving assignments and monitoring.
VT-10	Tamper-evidence check	Create at least three audit records, modify/remove a prior record in the test copy, then verify that integrity checking reports failure and identifies the affected record/chain point.
VT-11	Accessibility	Changing floor/direction, arrival, and emergency state are available as accessible text labels and a non-visual announcement in the prototype.
VT-12	Benchmark reproducibility (optional)	Run the same traffic profile twice with the same settings/seed where supported. Request-level and aggregate outputs are generated, and differences are explainable by recorded configuration or simulation randomness.
VT-13	Emergency override and rescue car	An authenticated Admin issues a permitted override and dispatches a rescue car to a stuck car: an eligible empty car other than E5 is chosen, its pending ordinary assignments are requeued, and the command, user, time, and outcome are audited. The same command from a passenger or technician is rejected, no command is sent, and the rejection is audited. For a medical report, the Admin sends a rescue car with a wait time, once before and once after the ill person boards: the car arrives, holds its doors open for that time without a stuck-door fault, and on Admin confirmation runs non-stop to the responder floor; on cancellation or expiry the car is released and the incident stays open.
VT-14	Freight elevator service run	No hall call or B1 request is assigned to E5. An authorized service run on E5 goes directly to its destination without stopping for resident calls; while it runs, E5 shows OCCUPIED and rejects new service runs and freight calls with a hall-panel message.
VT-15	Empty hall call (FR-1.21)	A car arrives for a hall call and no car call, load change, or obstruction event follows within 30 seconds. The request is COMPLETED, flagged NO_BOARDING, and its wait time is recorded.
VT-16	Duplicate calls and no cancel (FR-1.20, FR-1.22)	Pressing the same hall key five times while a request is active leaves one request. No passenger control cancels it. An Admin cancellation is recorded with actor and reason. Five NO_BOARDING requests from one floor within 10 minutes raise a low-severity alert, and calls are still served.
VT-17	Lost card (SEC-2.16, SEC-2.17)	A card reported lost is suspended and denied at a restricted floor, and raises an alert on the first attempt, while hall calls and travel to B1 still work. In one run the card is found within the window and reinstated after Admin identity confirmation. In another run the window ends and the card is revoked automatically. A replacement card gets a new identifier and the permission set only after Admin confirmation. All actions are audited.
VT-18	Cloned card (SEC-2.18)	One credential is presented in two cars within 10 seconds. The later request is denied, an alert names both cars, the credential becomes SUSPENDED, and the attempts are audited.
VT-19	Unregistered card (SEC-2.19)	An unknown identifier requests a restricted floor. The request is denied and counted as a failed attempt for that identifier. Three attempts within 60 seconds raise an alert.
VT-20	Medical emergency (EMR-3.1, EMR-3.14)	The medical option is triggered with no card. The car becomes EMERGENCY/UNAVAILABLE, the dashboard alert shows within 1 second with car, floor, and direction, hall calls are requeued, car calls are blocked, and the car runs

Test ID	Scenario	Pass criteria
		non-stop to the responder floor and opens its doors. With a fire active, the car does not run.
VT-21	Repeated emergency triggers (EMR-3.15)	Four triggers on one car within 60 seconds are all recorded as valid, and one low-severity alert links them. An Admin can close the incident as FALSE_ALARM, and the closure is audited.
VT-22	Sensor fault (HW-1, HW-2)	A stuck obstruction sensor keeps the close command inhibited and escalates after 10 seconds. A silent position sensor makes the car state stale, the sensor is marked FAULTY, and staff are notified.
VT-23	Traffic profiles (PERF-8 to PERF-10)	In the peak and off-peak profiles, the peak average wait is at most 60 seconds (95th percentile at most 120 seconds), the off-peak average wait is at most 90 seconds.
VT-24	Capacity and load (CAP-1, PERF-1, PERF-2, PERF-4, PERF-5)	With 600 apartments, 3,000 credentials, and 100 active requests on E1 to E5, feedback and state refresh stay within 1 second and no request is dropped. A backlog above 100 is shown on the dashboard.
VT-25	Backup and restore (DAT-2, DAT-3)	A backup is taken, records are added, and the data store is restored. The restore takes at most 1 hour, the data matches the backup, and the audit-integrity check passes.
VT-26	SREMS failure (MNT-4.11, AVL-1)	With SREMS stopped, no commands are issued. With the data store stopped and SREMS running, restricted destinations are denied (B1 and an active emergency policy excepted). A security decision that cannot be written to the audit log is rejected. Downtime is recorded for AVL-1.
VT-27	Escalation and concurrent incidents (EMR-3.9, EMR-3.13)	An incident left unresolved past the threshold re-notifies security and the escalation is recorded. With a fire and a stuck car active together, the fire restrictions apply and the stuck car incident stays open.
VT-28	External message validation (SEC-2.10, Section 3.3.3)	Messages with an unknown elevator, floor 31, an unknown type, or an illegal state transition are each rejected and logged, and the last valid state is kept.
VT-29	Search and export (SEC-2.14, ANA-5.7, ANA-5.9)	Incident and audit records can be searched by time range, card, car, event type, and status, and the results match the stored records. Per-request timestamps and aggregates export for a recorded run.
VT-30	HELP request (EMR-3.1, Table 5)	A HELP trigger creates an incident, a dashboard alert, and an in-car message, and the car stays eligible for dispatch. Five HELP presses on one car do not remove it from dispatch. When the equipment reports a stuck event, or an Admin confirms the HELP request, the car becomes EMERGENCY/UNAVAILABLE and its hall calls are requeued.
VT-31	One presentation and visitor pass (SEC-2.21, SEC-2.22)	After one card presentation, a second restricted destination selection is denied. A visitor pass opens only its named floors, stops working at its expiry time, and every issue and expiry is audited.
VT-32	Card scanning and command conflict (SEC-2.19, SEC-2.20)	Eleven distinct unregistered identifiers within 5 minutes raise one scanning alert. A rescue dispatch for a car that another Admin has just put into maintenance is rejected with a reason and audited.

Validation order: confirm the elevator-interface capabilities (OR-01); implement one state update, hall call, and car call; test access/B1 policy, one emergency path, emergency override, communication loss, maintenance isolation, audit integrity, accessibility, and, if time allows, a repeatable benchmark. Review results with all six team members, resolve contradictions, and accept the implementation only when the core scenarios have repeatable pass/fail outcomes.

Table 10: Traceability matrix

Need ID	Stakeholder need	Use cases	Requirements	Tests
SN-01	Passenger: short waiting time	UC-01	FR-1.3 to FR-1.7, FR-1.18, FR-1.23, PERF-8 to PERF-10, ANA-5.2	VT-01, VT-05, VT-23
SN-02	Passenger: reach a permitted destination	UC-02, UC-08	FR-1.8 to FR-1.10, SEC-2.1 to SEC-2.6	VT-02, VT-03, VT-17
SN-03	Passenger: predictable door, load, and status behavior	UC-02, UC-15	FR-1.2, FR-1.11, FR-1.14 to FR-1.17, EMR-3.10, HW-1, HW-2	VT-06, VT-22
SN-04	Passenger: emergency help and accessible information	UC-04, UC-10, UC-14	EMR-3.1 to EMR-3.7, EMR-3.11, EMR-3.14, EMR-3.15, UI-6, UI-7	VT-07, VT-11, VT-20, VT-21, VT-30
SN-05	Building owner: capacity and continuity	UC-01, UC-09	FR-1.6, PERF-4, PERF-5, CAP-1, AVL-1, MNT-4.8 to MNT-4.11, DAT-1 to DAT-4	VT-08, VT-24, VT-25, VT-26
SN-06	Building owner: low downtime and maintenance burden	UC-05, UC-06	MNT-4.1 to MNT-4.7, ANA-5.5, AVL-1	VT-09, VT-26
SN-07	Security manager: restricted-floor access	UC-02, UC-03, UC-08, UC-13	SEC-2.1 to SEC-2.7, SEC-2.9, SEC-2.10, SEC-2.12, SEC-2.13, SEC-2.15 to SEC-2.22	VT-02, VT-04, VT-10, VT-17, VT-18, VT-19, VT-28, VT-31, VT-32
SN-08	Security manager: awareness of suspicious and emergency activity	UC-06, UC-10, UC-11, UC-16	SEC-2.7, SEC-2.8, SEC-2.11, SEC-2.14, FR-1.20 to FR-1.22, EMR-3.9, EMR-3.13, ANA-5.1, ANA-5.6 to ANA-5.8	VT-04, VT-10, VT-15, VT-16, VT-21, VT-27, VT-29
SN-09	Technician: safe maintenance access and fast diagnosis	UC-05, UC-06	MNT-4.3 to MNT-4.7, UI-4	VT-09
SN-10	Building-service staff: freight and service operation	UC-12	FR-1.13, SEC-2.2	VT-14
SN-11	Security manager: emergency override and rescue	UC-07	EMR-3.8, EMR-3.12, EMR-3.16	VT-13

Table 11: Non-functional requirements matrix

NFR ID	Attribute	Requirement	Metric and target	Verification	Tests
NFR-01	Performance	PERF-1	Visible feedback for ordinary actions: at most 1 s	Timed UI test	VT-24
NFR-02	Performance	PERF-2	State shown on the dashboard: at most 1 s	Timed test	VT-24
NFR-03	Performance	PERF-3	Critical alert on the dashboard: at most 1 s	Timed test	VT-07, VT-20

NFR ID	Attribute	Requirement	Metric and target	Verification	Tests
NFR-04	Performance	PERF-9	Peak: average wait at most 60 s, 95th percentile at most 120 s	Peak profile run	VT-23
NFR-05	Performance	PERF-10	Off-peak: average wait at most 90 s	Off-peak profile run	VT-23
NFR-06	Capacity	CAP-1	600 apartments, 3,000 credentials, 100 active requests	Load run	VT-24
NFR-07	Capacity	PERF-4, PERF-5	Five cars served; backlog limit 100 active requests; no request dropped	Load run	VT-24
NFR-08	Availability	AVL-1	At least 99.5% per month (about 3.6 h of downtime)	SREMS restart log and heartbeat check	VT-26
NFR-09	Availability	Table 7	A fault in one car never disables unrelated cars	Fault injection	VT-09
NFR-10	Data retention	DAT-2	Records kept at least 12 months; audit records archived, not deleted	Inspection and restore test	VT-25
NFR-11	Recovery	DAT-3	Backup every 24 h; audit records copied as written; restore at most 1 h; at most 24 h of non-audit records lost	Backup and restore test	VT-25
NFR-12	Data protection	DAT-4, SEC-2.13	Role checks on all data access; no biometric data stored	Inspection and role test	VT-02
NFR-13	Security	SEC-2.6	100% of unauthorized destination requests blocked before any command	Security test suite	VT-02, VT-17, VT-19
NFR-14	Security	SEC-2.7, SEC-2.18	Alert at 3 failures in 60 s; alert on a cloned credential	Alert tests	VT-04, VT-18
NFR-15	Integrity	SEC-2.12, ANA-5.8	100% of modified or removed audit records detected; first bad record reported	Tamper test	VT-10, VT-25
NFR-16	Reliability	COM-4, MNT-4.9	No blind retry; no duplicate execution; state reconciled before control resumes	Fault injection	VT-08
NFR-17	Reliability	MNT-4.11	Restricted destinations denied when the data store is down	Fault injection	VT-26
NFR-18	Accessibility	UI-6	Floor, direction, arrival, and emergency state as text and non-visual announcement	Accessibility test	VT-11
NFR-19	Testability	Section 3.5, SWI-2	All core behavior testable with labelled test stubs	Inspection	All VT

Table 12: Revision history

Version	Date	Description
0.1	October 7, 2026	Draft outline and first requirements.

Version	Date	Description
0.2	October 9, 2026	Team review revision: direction-aware dispatch, request lifecycles, incident-response table.
1.0	October 12, 2026	Final submission: elevator interface, rescue car, freight service run, fail-closed rule.
1.1	October 19, 2026	Added call-abuse and empty-call rules, credential lifecycle, medical emergency, sensors and interface requirements, traffic profiles, capacity, availability and data requirements, new use cases UC-10 to UC-16, tests VT-15 to VT-32, the separate freight call, lost-card suspension, medical rescue wait, and Tables 10 to 12. Final review: SREMS-down case, freight-call rejection, priorities, and traceability notes.

5 References

ISO/IEC/IEEE. (2018). ISO/IEC/IEEE 29148:2018, Systems and software engineering - Life cycle processes - Requirements engineering. International Organization for Standardization.

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